ROOT CAUSE ANALYSIS OF ISSUES IN ROCKET LAUNCHING SYSTEM

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MAY, 2020

CERTIFICATE

This is to certify that the project titled **"ROOT CAUSE ANALYSIS OF ISSUES IN ROCKET LAUNCHING SYSTEM"** is a bonafide record of the work done by

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In partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology** in **Mechanical Engineering** of the Gayatri Vidya Parishad College of Engineering (Autonomous) affiliated to Jawaharlal Nehru Technological University, Kakinada during the year 2016-2020.

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ABSTRACT

Decoys in navy applications are usually the containers of aluminum particles used to mislead the incoming enemy missiles by forming an aluminum cloud. Proper rocket launching system of decoys is very important in order to protect the vessel. Kavach rocket system of Indian Navy has occasional malfunctions such as rocket not leaving the launcher and rocket leaving the launcher without tail fins. In this project, root cause analysis for these malfunctions will be carried out and appropriate solutions will be proposed. Possible causes such as insufficient thrust, high strength shear pin, thermal mismatch between rocket and launcher tube will be investigated. Entire rocket launching system will be modeled in CATIA and analyzed in ANSYS to find out the root cause and validate the recommended solutions.

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LIST OF ABBREVATIONS

MRCR	Medium Range Combat Rocket
CNAI	Controllerate of Naval Armament Inspection
RPN	Risk Priority Number
FEM	Finite Element Model
FEA	Finite Element Analysis
CAD	Computer Aided Drawing

CHAPTER 1 INTRODUCTION

Kavach is an anti-missile naval decoy system to distract radar guided missiles from their targets and act as a system for self-defense. The Kavach system releases chaff made up of aluminum particles. The chaff forms a cluster which remains suspended in the air so that the incoming guided missile continues towards the chaff as the actual target and gets locked on to the chaff instead of actual target. A typical Kavach rocket launching system is shown in Fig 1.1

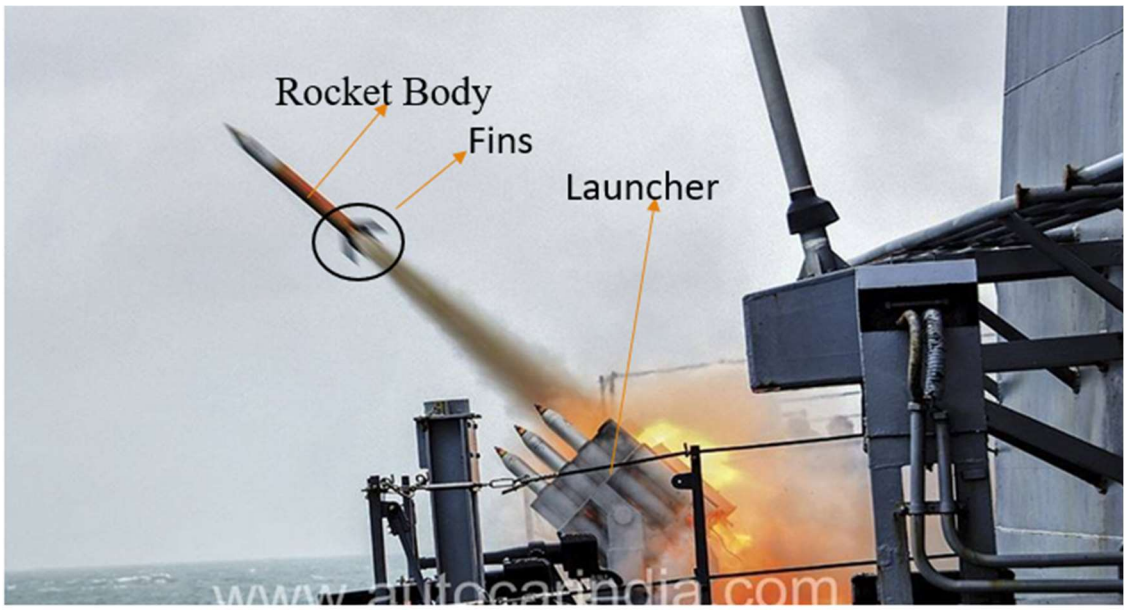


Fig 1.1 Kavach Rocket launching system [1]

The Kavach system has Decoys of three different versions based on the range [2]

- Long Range: 12 km (7.5miles)
- Medium Range: 5 km (3.1 miles)
- Short Range: 1 km (0.62miles)

These versions are fired in different situations based on incoming threats. The first ship to be equipped with Kavach was INS Kamorta. Now it is on-board all the vessels of the Indian Navy such as:

- INS Kiltan
- INS Kavaratti
- INS Betwa

- INS Chennai
- Kolkata-class destroyers
- Visakhapatnam-class destroyers
- Shivalik-class Fighters

Kavach launching system of Indian Navy has occasional issues such as:

1. Rocket not leaving the launcher tube
2. Rocket leaving the launcher without tail fins.

1.1 Consequence of issue 1

If anyone of the rocket gets struck in the launcher assembly, clutter formed by remaining decoys becomes incapable of misguiding the incoming enemy missiles which harm the vessel.

1.2 Consequence of issue 2

The purpose of putting fins on a rocket is to provide stability during its flight, that is, to allow the rocket to maintain its orientation and intended flight path. If a rocket was launched without fins, it would soon tumble after leaving the launcher, due to the way that aerodynamic and other forces (such as wind) act upon the rocket.

In this project, root cause analysis for these issues of MRCR will be carried out and solutions will be proposed.

1.3 Introduction to Root Cause Analysis

Root Cause Analysis [2] is a method that is used to address a problem or non-conformance, in order to get the “root cause” of the problem. It is used so that, the cause can be corrected or eliminated, and the problem can be prevented from recurring. Root cause analysis is simply the application of a series of well known, common sense techniques which can produce a systematic, quantified and documented approach to identification, understanding and resolution of underlying causes. The process flow in root cause analysis is as as shown in Fig 1.2

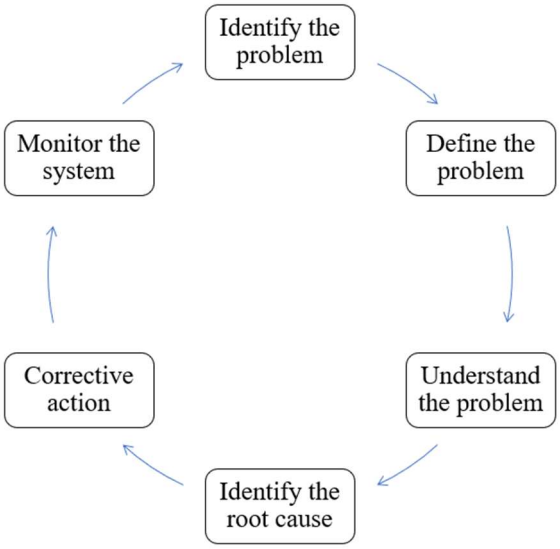


Fig 1.2 Process flow in Root Cause Analysis

1.4 Root Cause Analysis Tools & Techniques

- 5 Why's Method
- Pareto Chart
- Scatter Diagram
- Failure Mode and Effects Analysis
- Fishbone Diagram

1.4.1 5 whys method

5 Whys in an iterative interrogative technique used to explore the cause and effect relationships underlying a particular problem. The primary goal of the technique is to determine the root cause of a defect by repeating the question "Why?". Each answer forms the basis of next question.

1.4.2 Pareto chart

A Pareto chart is a histogram or bar chart combined with a line graph that groups the frequency of different problems to show their relative significance. The bars shows the frequency in descending order, while the line shows cumulative percentage or total as moved from left to right.

1.4.3 Scatter Diagram

A Scatter plot diagram uses pairs of data points to help uncover relationship between variables. A Scatter plot is a quantitative method for determining whether two variables are correlated. Making Scatter diagram is as simple as plotting independent variable (or suspected cause) on the x axis and dependent variable (the effect) on y axis. If the pattern shows a clear line or curve then the variables are correlated and can proceed to correlation analysis.

1.4.4 Failure Mode and Effects Analysis

It is used during product or process design to explore potential defects or failures. An FMEA chart outlines:

- Potential failures, consequences and causes.
- Current controls to prevent each type of failure.
- Severity, Occurrence, and detection ratings that allow the calculation of risk priority number (RPN) for determining further action.

1.4.5 Fishbone Diagram

It is also known as Ishikawa diagram or cause and effect diagram. This is useful technique that is used for more complex Root cause analysis. This type of diagram identifies all the potential processes and factors that could contribute a problem[4]. A basic Fishbone diagram is shown in Fig 1.3.

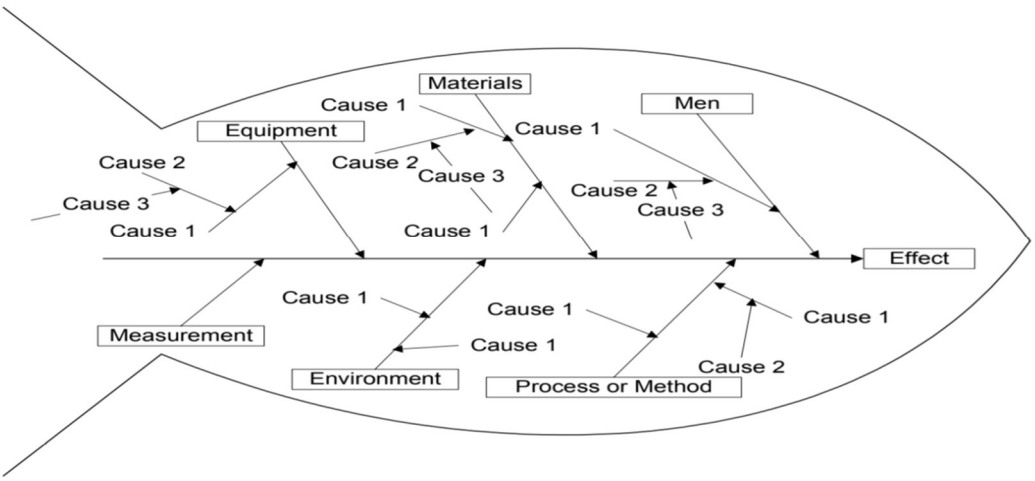


Fig 1.3 Basic Fishbone diagram

Out of all the Root cause analysis Methods Fishbone diagram has the following benefits so this method is adopted to find out the root cause in this project.

1.5 Benefits of Fishbone diagram

- It is easy to understand and analyze.
- It involves an in-depth discussion of the problem.
- It helps to locate bottlenecks in the process.
- It prioritizes further analysis and helps to take corrective action.

1.6 Fishbone Diagrams for Issue-1 & Issue-2

1.6.1 Fishbone Diagram for Issue-1

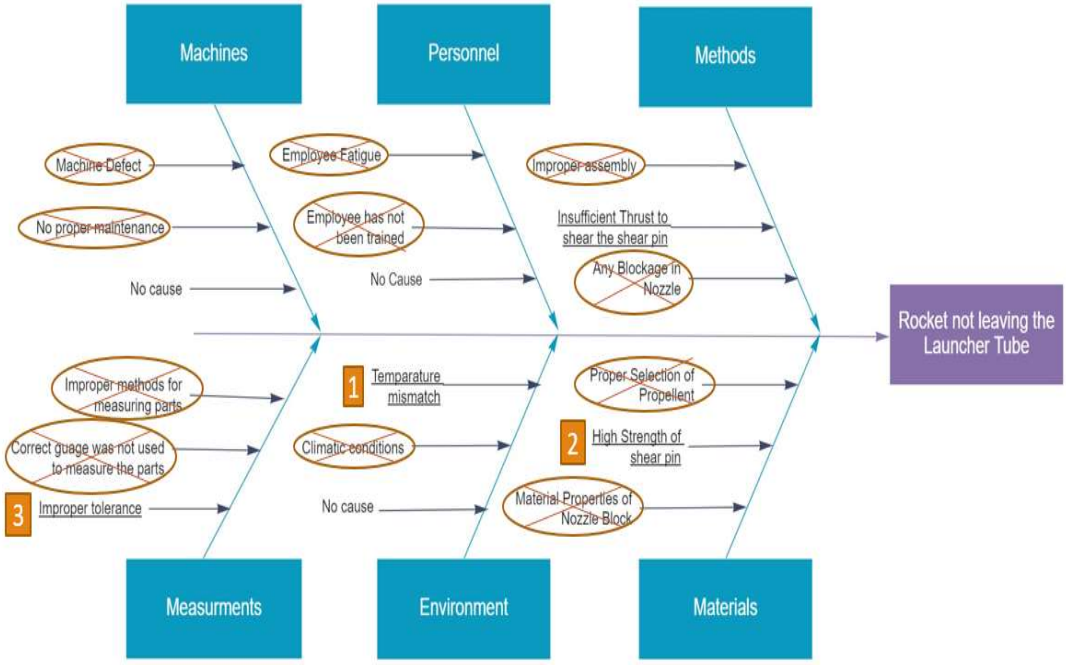


Fig 1.4 Fishbone diagram for issue-1

For the above discussed problem, the possible causes are brainstormed.

i. **Temperature mismatch:** The diametrical gap between rocket and launcher tube is 0.5mm. Here, high temperatures are encountered in the motor tube due to the combustion of propellant. As different materials exhibit different expansion characteristics. During combustion, if major diameter of rocket exceeds the minor diameter of launcher tube, rocket may get stuck in launcher tube.

ii. **High strength of shear pin:** Though sufficient thrust is developed by proper selection of propellant, if the shear strength of shear pin is so high such that rocket is unable to shear the shear pin, then the rocket gets stuck in the launcher tube.

iii. **Proper selection of propellant:** If proper propellant is not selected, combustion of propellant will be non-uniform and thrust generated will inappropriate to shear the shear pin.

iv. **Material Properties of Nozzle Block:** If the material properties of nozzle block are such that, though there is sufficient thrust present nozzle block groove is unable to shear the shear pin.

v. **Any blockage in nozzle:** Due to the presence of any dust particles in the nozzles may affect the thrust to be generated.

By the information given by CNAI, it is concluded that causes like selection of propellant, blockage in nozzle can be ignored as they maintained quality loops before assembling the rocket assembly. Here, the causes like temperature mismatch due to thermal deformation and strength of the shear pin are focused.

1.6.2 Fishbone Diagram for issue-2

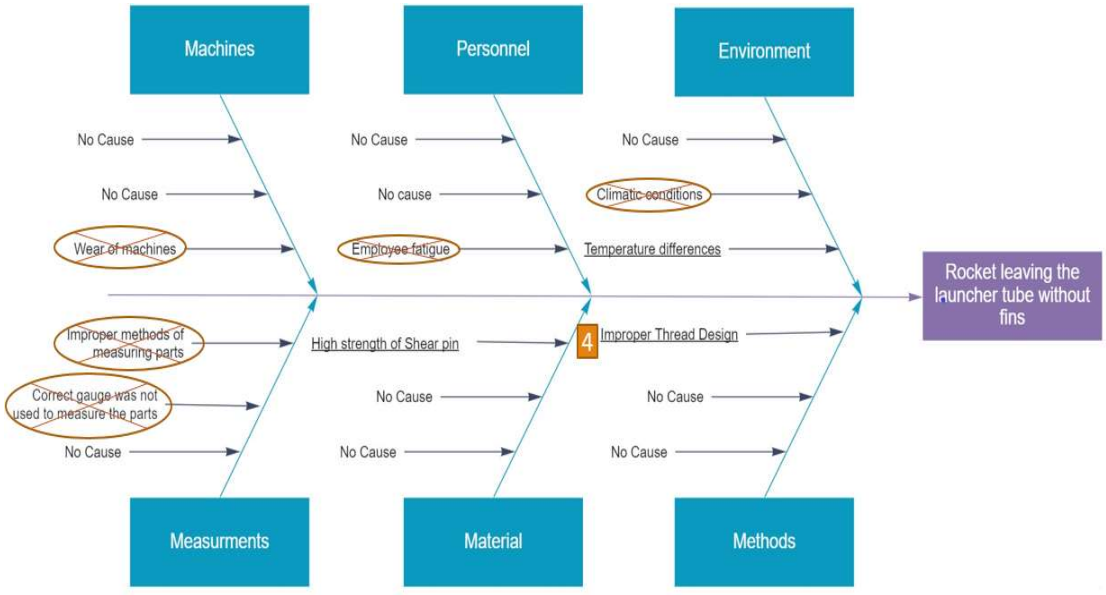


Fig 1.5 Fishbone Diagram for issue-2

For the above discussed problem, all the possible causes are brainstormed:

i. **Improper thread design:** If the thread joint between motor tube and nozzle block has low strength such that rocket may move forward leaving the fins behind during launching.

ii. **High Strength of shear pin:** If shear strength of the shear pin is so high such that though the thrust generated is able to shear the shear pin, it reduces the strength of the thread joint between motor tube and nozzle block.

iii. **Temperature differences:** If the thermal deformation of fins and launcher tube may be such that it may lead to blocking of fins in the launcher tube and rocket moving ahead without fins.

CHAPTER 2 LITERATURE REVIEW

Viveros et al. [1] Failure causes of production systems need to be identified, for effective recovery solutions. Root cause analysis can be used for this, integrated by a method for identifying coherent and sustainable solutions. In this paper, root cause analysis for failure cause identification is coupled with a method of problem solving to guide the search for solutions, called theory of inventive problem solving. The integration of these methodologies is implemented and demonstrated on a real case concerning the solution to the potential failure of the equipment used in the torque test performed on the truck engines used in high-tonnage mining operation. The incorporation of a key performance indicator analysis allows evaluating realistically the results obtained and determining the success of the solution.

Geoff Vorley [2] investigated root cause analysis of several cases in manufacturing industries, suggested the methods and procedures to be followed in his book "Mini Guide to Root Cause Analysis". He discussed root cause analysis significance and mentioned different tool and techniques such as 5WHY'S, Pareto analysis, Fish bone diagrams, Cause and Effect diagrams, Process analysis, Mapping and Flow charts, Fault tree, check sheets.

Anoop Thankachen et al [3] investigated on the design optimization and stress analysis of rocket structure for aerospace application. Stress analysis plays important role in the structural design and safety. In this paper, a test rocket structure is designed for aerodynamic loads. The test rocket comprises main parts of Nose, body, tail and Fins. Firstly, they designed these parts and assembled it later. Theoretical calculations are carried out for the sectional requirement of the various members and later the overall rocket structure is design optimized using Ansys design optimizer. The model is then imported into hyper mesh software in order to mesh the model properly and then analysis is conducted in ANSYS software.

Shahneen et al. [4] investigated on the design and thermal analysis of rocket motor casing. In this paper, the design of the rocket motor case is done by considering the maximum expected operating pressure, design safety factor of 1.25 with diameter of 11 inches and thickness of 0.06 inches. The 3-D model of rocket motor case is developed

using CATIA V5R16 software. The static structural analysis and linear buckling analysis is done for unidirectional carbon-epoxy IM10/8552 composite and the steady-state thermal analysis is done on carbon-epoxy IM10/8552 shell model and solid model of the rocket motor casing by using ANSYS 15.0 software and the results are compared with the results of static structural, steady-state thermal, linear buckling analysis of D6AC steel material rocket motor casing to specify the better efficient material.

Mansuy et al [5] performed a new calculation method for worst case tolerance analysis and synthesis in stack type assemblies. As name suggest new approach is considered for the stack assemblies without clearance and with clearance. It is concluded that there are so many factors that affect the stack-up analysis which assembly having clearance. It is also concluded that any kind of complex assembly can be converted into a single chain assembly with the help of this research work and can make relationship between functional condition and geometric tolerance by considering worst case. To make an assembly in a single chain influence coefficient is used which direct the chain and decide whether to make summation or subtraction of tolerances. Among all these influence coefficients which have the higher value is considered as reference.

Srinivasa Rao et al [6] analyzed the failure mode of the deep groove ball bearing consisting of outer ring, inner ring, balls, and the cage. In the paper, 6200 deep groove ball Bearing is considered and four different materials are used for analysis. The deep groove ball Bearing is modeled by using CATIA. Stresses, deformations and strains for varying applied loads are obtained from static structural analysis by using ANSYS. The variation in the deformations and the stresses with respect to time due to impact loading are studied by using transient dynamic loading using ANSYS.

Kiran Kumar et al [7] the most common failure method for pins is through shear failure. In the present study, design of closing blade pin is carried out to study the shear strength of the pin material during blade off situations in aero engines. Experimental testing of pin shear was carried out; finite element analysis of pin shear phenomenon was studied using ANSYS 14.5 version. The analysis was carried out for full loading conditions for various materials such as stainless steel, aluminum and titanium. Results obtained from ANSYS were validated against experimental data and comparative details are presented.

Zhang et al [8] studied effects of geometric parameters in fracture test, this paper used 3D modeling software Solid works and finite element analysis software Deform to simulate the fracture process of rocket shear pin and study the effects on shear test while various geometric parameters change in the assembly of fixtures and shear pins.

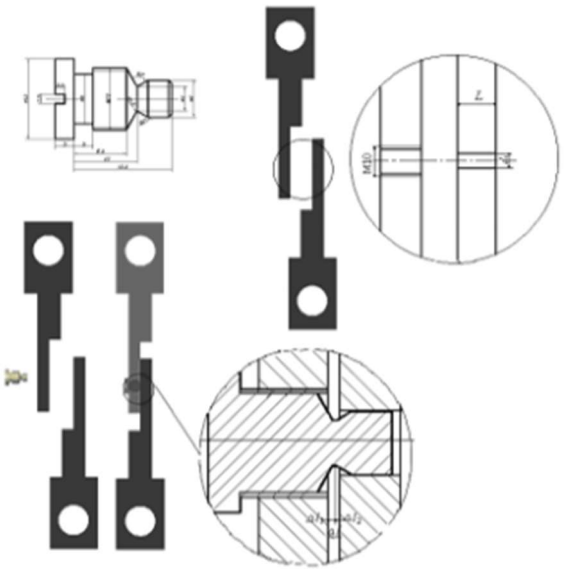


Fig 2.1 Shear pin and its fixtures

Pawar et al [9] studied on computer aided modeling of components of bolted joint and their finite element analysis has been carried out through ANSYS workbench software under definite loading conditions, to obtain different stress and displacement values. The analysis includes detailed stress analysis of a single lap bolted joint. The spatial variation of total deformation, directional deformation, equivalent stresses and principal stresses.

Reddy et al[10] examined the effects of the fin shape of the heat sink on the thermal performance. The various types of Perforation on pin fin are used for effective heat transfer rate under constant heat flux condition. The numerical simulation is accomplished by 3D modeling and analysis using CATIA and ANSYS.

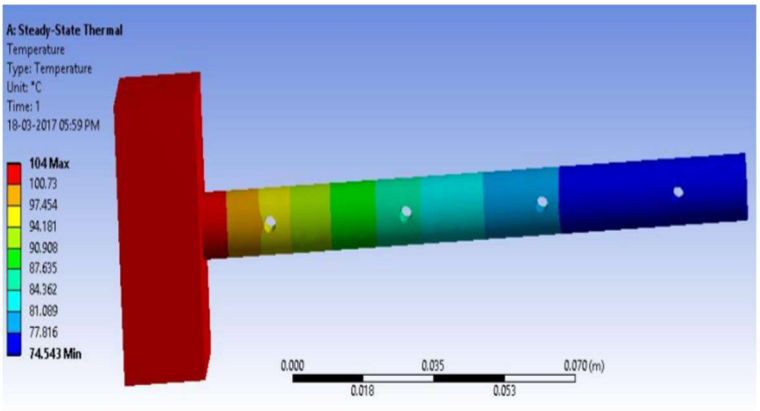


Fig 2.2 Temperature distribution of pin fin

The main goal is to increase the heat transfer rate through the fin surface and to decrease material cost**.** The effect of different shape forms of pin fins on the overall performance of the heat sink studied in this paper.

CHAPTER 3 MODEL PREPARATION

Modeling provides the design engineer comfortable modeling, and dimension driven editing. An excellent way to begin a design concept is with sketch. When a sketch is used, a rough idea of the part is represented and constrained based on the fit and function requirements of the design. In this way, the design intent is captured. This ensures when the is passed to the next level of engineering, the basic requirements are not lost when the design is edited.

Kavach Rocket Launching Assembly contains various parts. They are

1. Fuse adapter
2. Body
3. Base Adapter
4. Motor tube
5. Nozzle Block
6. Fins
7. Launcher tube
8. Stopper Pin
9. Shear pin
10. Central and Periphery Nozzles

3.1 Basic Introduction about the Kavach parts

1	**Fuse Adapter**	:	It consists of electrical setup to which is used to set the spark.
2	**Body**	:	It consists of aluminum chaff which forms the clutter.
3	**Base adapter**	:	It connects the body and motor tube in the rocket. It also consists of small RDX material which is ignited to combust the solid propellant.
4	**Motor Tube**	:	It consists of solid propellant which combusts to give necessary thrust to the rocket.

5	**Nozzle Block**	:	It consists 6 peripheral nozzles and 1 central nozzle through which the exhaust gases escape.
6	**Fins**	:	These are attached to the nozzle block grooves to give stability to the rocket.
7	**Launcher tube**	:	The whole rocket assembly is placed inside the launcher tube. It consists of two holes at certain distance for the placement of stopper pin and shear pin.
8	**Stopper pin**	:	It is fitted to the launcher tube through hole such that the entire rocket rests on the tip of the stopper pin.
9	**Shear pin**	:	Once the rocket is placed in the launcher tube it is placed in the hole present on the launcher tube such that its tip sits in the nozzle groove.
10	**Central& Periphery Nozzles**	:	A Rocket Nozzle is in the purpose of expanding and to accelerate the combustion gases produced by burning propellants So that exhaust gases exit the nozzles at higher velocities There are six peripheries and one central nozzle in Kavach rocket system.

3.2 Solid Model Preparation

3.2.1 Introduction to CATIA

CATIA is acronym for Computer **A**ided **T**hree-Dimensional **I**nteractive **A**pplication. It is one of the leading 3D software used by organizations in multiple industries ranging from aerospace, automobile to consumer products.

CATIA is a multiplatform 3D software suite developed by Dassault Systems, encompassing CAD, CAM as well as CAE. Dassault is a French engineering giant active in the field of aviation, 3D design, 3D digital mock-ups, and product lifecycle management (PLM) software. CATIA is a solid modeling tool that unites the 3D parametric features with 2D tools and also addresses every design-to-manufacturing process. In addition to creating solid models and assemblies, CATIA also provides generating orthographic, section, auxiliary, isometric or detailed 2D drawing views. It is also possible to generate model dimensions and create reference dimensions in the

drawing views. The bi-directionally associative property of CATIA ensures that the modifications made in the model are reflected in the drawing views and vice-versa.

All the 3-D parts of rocket launcher assembly are designed and assembled in CATIA V5 software, according to the dimensions and assembly drawings given by CNAI.

3.2.2 CAD Models

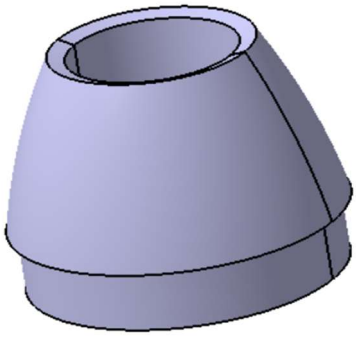


Fig 3.1 Adapter for Fuse

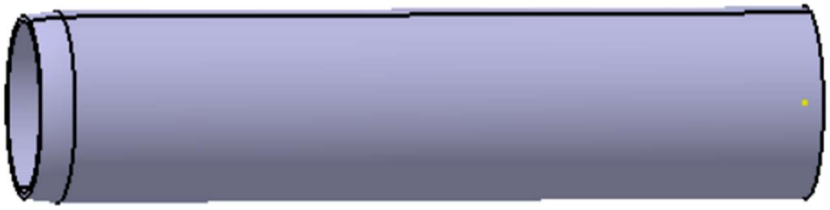


Fig 3.2 Body

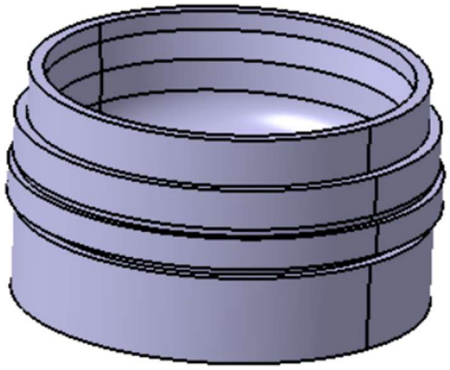


Fig 3.3 Base Adapter

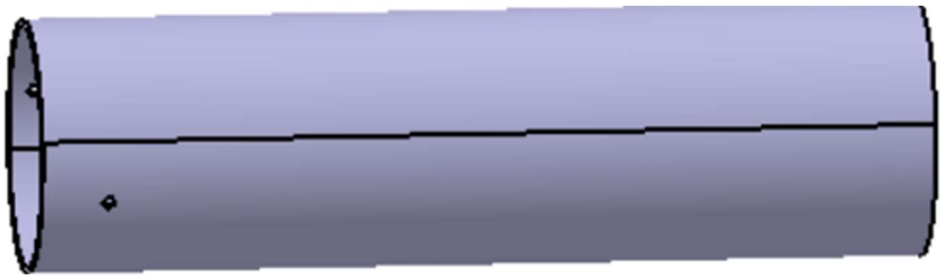


Fig 3.4 Motor Tube

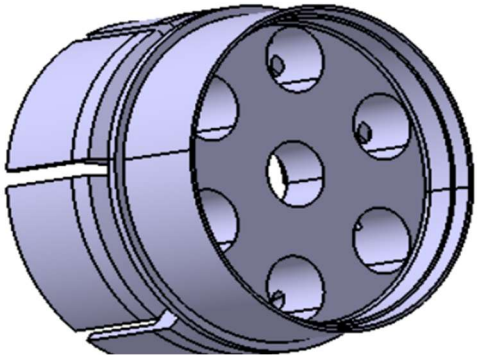


Fig 3.5 Nozzle Block

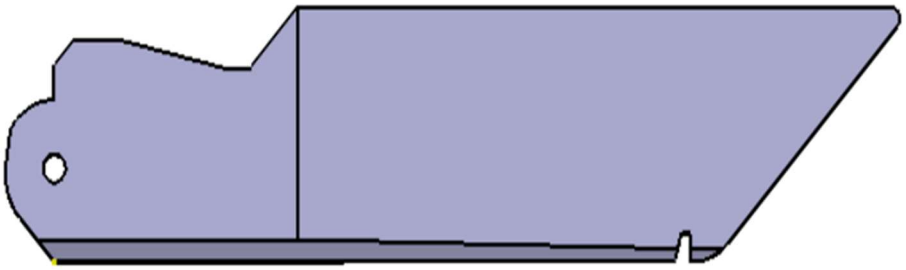


Fig 3.6 Fin

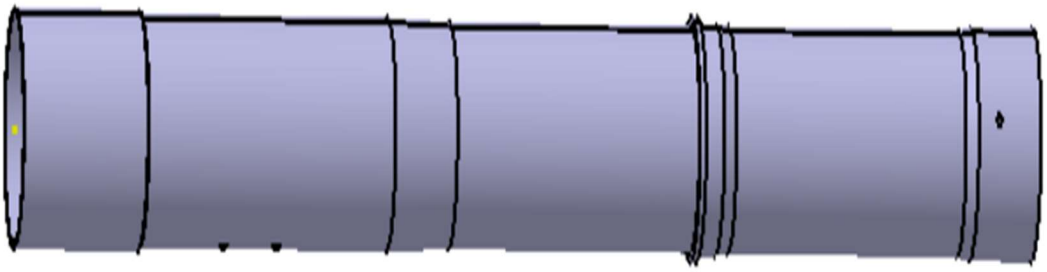


Fig 3.7 Launcher Tube

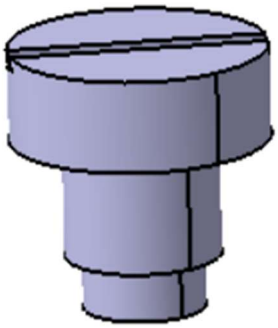


Fig 3.8 Stopper Pin

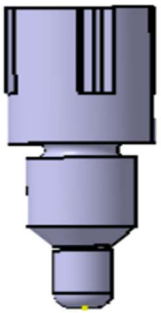


Fig 3.9 Shear Pin

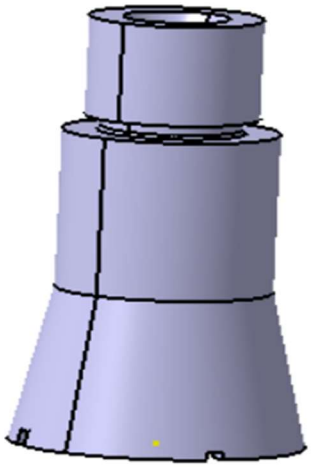


Fig 3.10 Central Nozzle

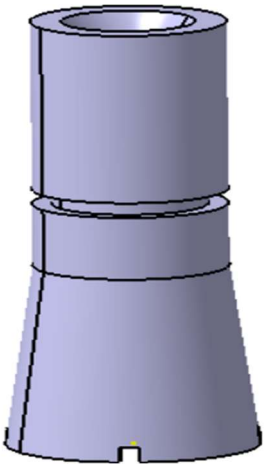


Fig 3.11 Periphery Nozzle

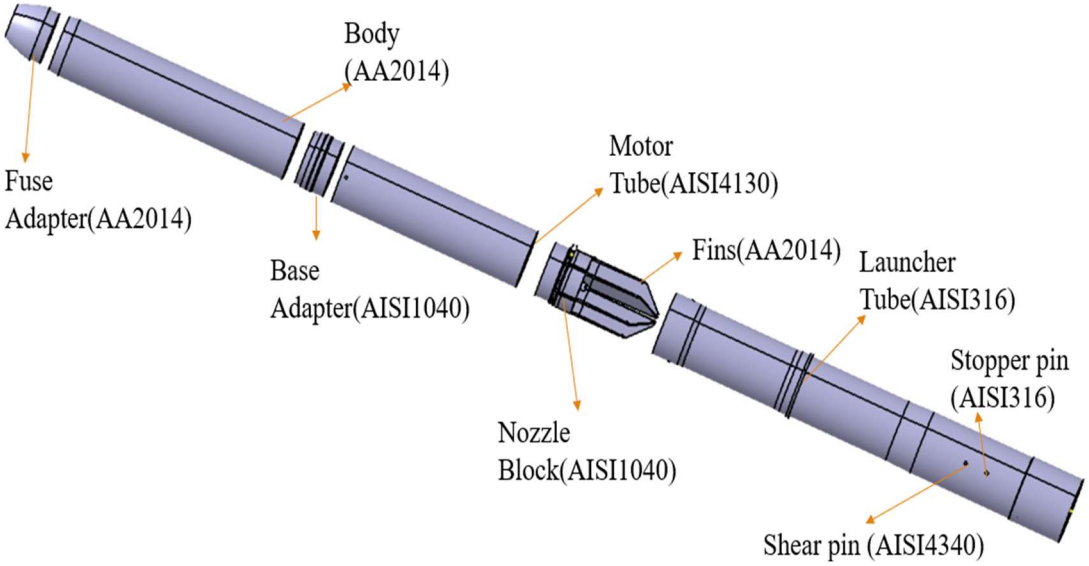


Fig 3.12 Rocket and launcher Assembly

3.3 Finite Element Model Preparation

The FEM is a mathematical tool for solving ordinary and partial differential equations. Because it is a numerical tool, it has the ability to solve the complex problem that can be represented in differential equations form. The applications of FEM are listless as regards the solution of practical design problems. Due to high cost of computing power of years gone by, FEA has a history of being used to solve complex and cost critical problems. Classical methods alone usually cannot provide adequate information to determine the safe working limits of a major civil engineering construction or an automobile or aircraft[5]

3.3.1 Steps involved in Finite Element Model Preparation

a) Solid Modeling in CATIA
b) Add Engineering data in ANSYS
c) Import geometry
d) Modeling
e) Contact generation
f) Mesh generation

a. Solid Modeling in CATIA

All the 3-D parts of rocket launcher assembly are designed and assembled in CATIA V5 software, according to the dimensions and assembly drawings given by CNAI.

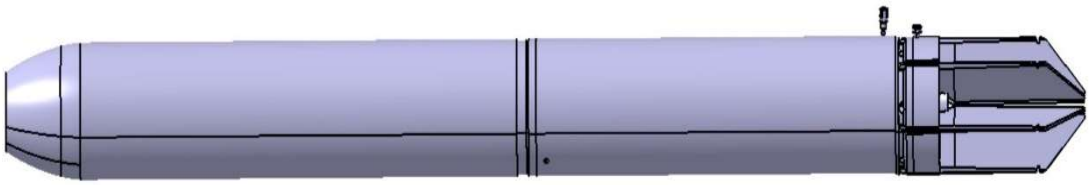


Fig 3.13 Rocket Assembly

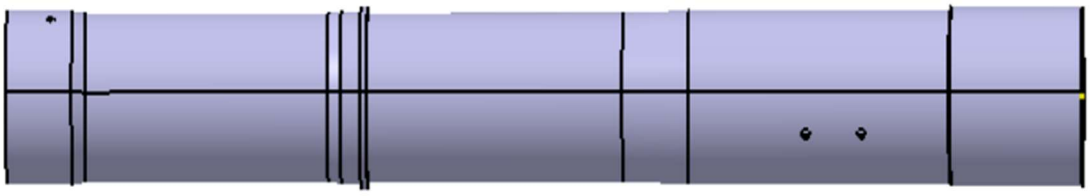


Fig 3.14 Launcher Tube

b. Add Engineering data in ANSYS

Engineering Data is a resource for material properties used in an analysis system. The Engineering Data workspace is designed to allow to create, save, and retrieve material models, as well as to create libraries of data that can be saved and used in subsequent projects.

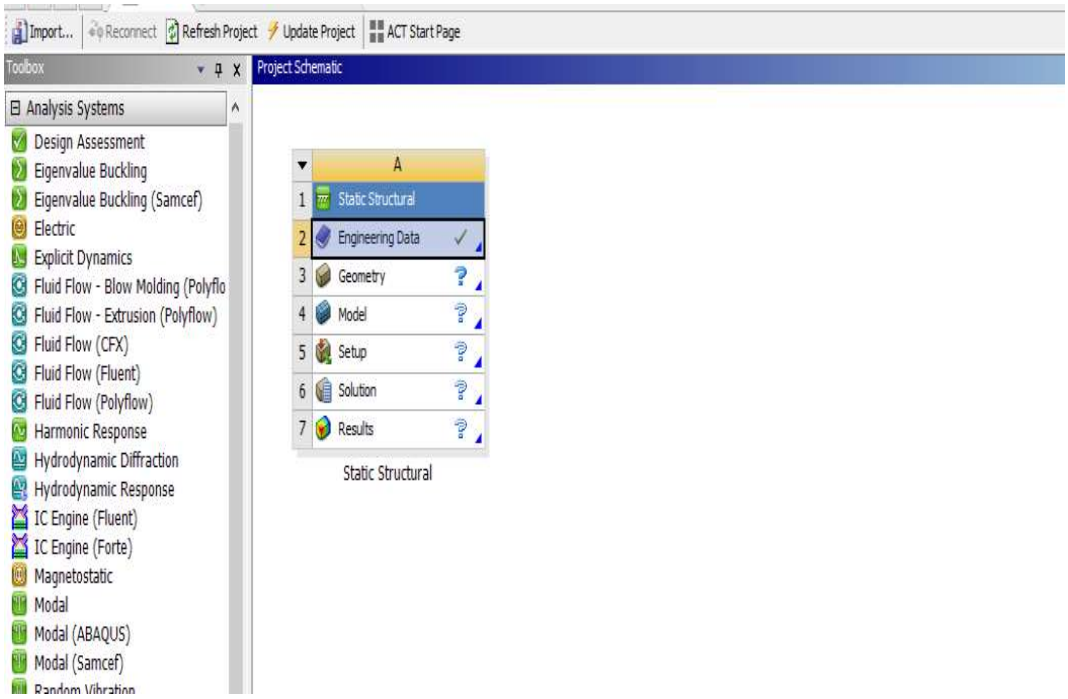


Fig 3.15 Layout of Project schematic in ANSYS Workbench

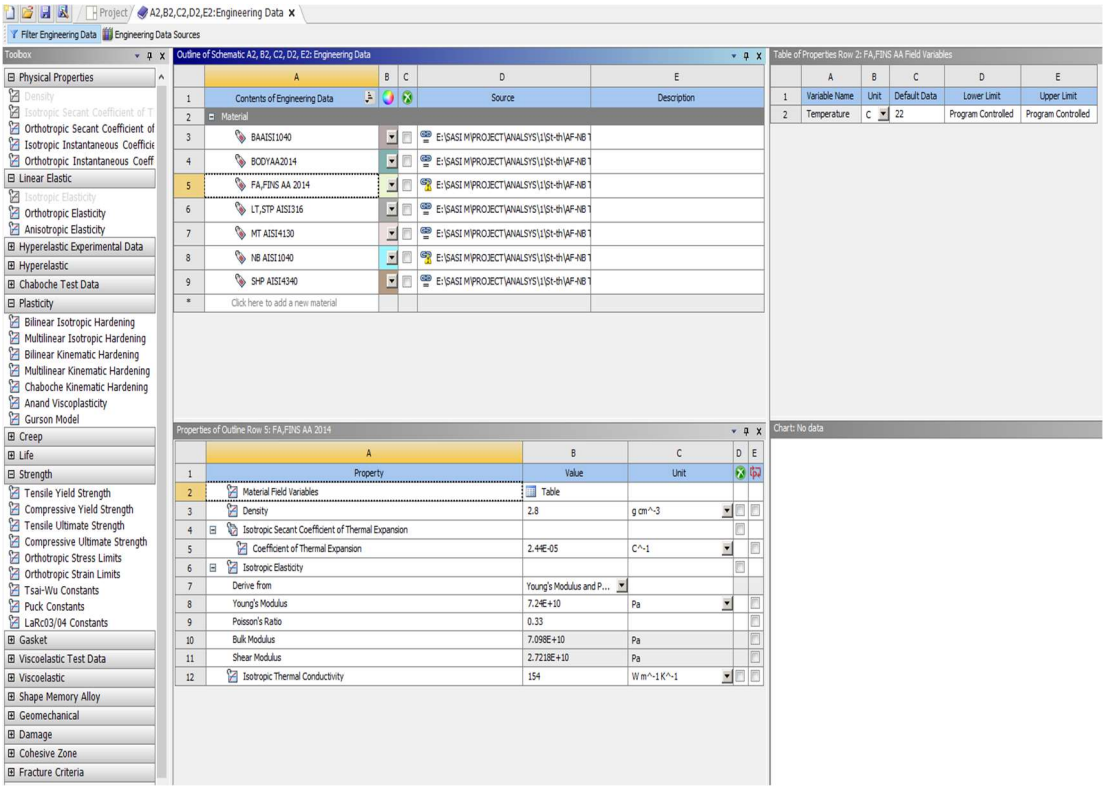


Fig 3.16 Outline of Schematic Engineering Data

3.3.1.1 Material Properties used in Analysis

Table 3.1 Properties of Materials

Properties	**Name of Material**				
	AA 2014 (Aluminum Alloy)	AISI 1040 (Alloy Steel)	AISI 4130 (Alloy Steel)	AISI 316 (Alloy Steel)	AISI 4340 (Alloy Steel)
Density(g/cm$^{3)}$	7.9417	5.0001	7.85	8.00	7.85
Coefficient of thermal Expansion(c^{-1})	2.44x10^{-5}	1..22x10^{-5}	1.17x10^{-5}	1.75x10^{-5}	1.38x10^{-5}
Young's Modulus (Pa)	7.24x10^{10}	20x10^{10}	20.5x10^{10}	19.3x10^{10}	20.5x10^{10}
Poisson's Ratio	0.33	0.29	0.29	0.30	0.29

c. Import geometry

After adding Engineering data the 3D Model designed in CATIA is either converted to step file or its files and it is imported to design modeler. The Design Modeler interface is shown in Fig 3.17

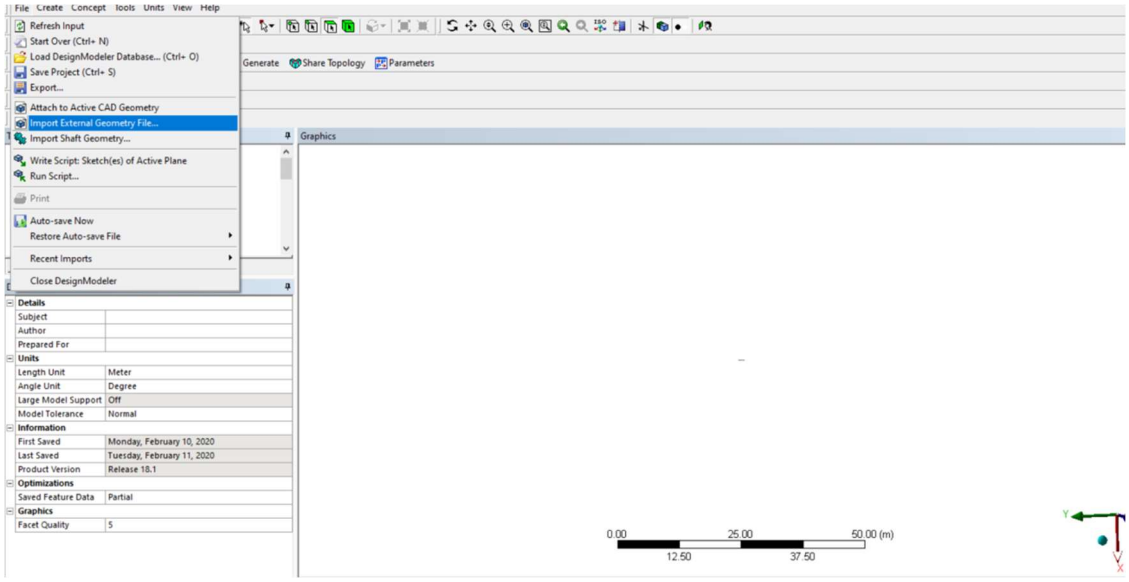


Fig 3.17 Design Modeler Interface

d. Modeling

The ANSYS Design Modeler application is designed to be used as a geometry editor of existing CAD models. The ANSYS Design Modeler application is a parametric feature-based solid modeler designed so that 2D sketches, modeling 3D parts, or uploading 3D CAD models can be drawn for engineering analysis preprocessing[7].

As the Launcher tube is in inclined position on the ship, the rocket inserted in the launcher tube rocket will get seated by touching the inner bottom section of the launcher tube. The sectional view of assembly is shown in Fig 3.18

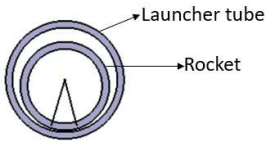


Fig 3.18 Sectional view of Rocket position in Launcher

The rocket is inserted in the launcher tube rests on the stopper pin as the length of rocket assembly is greater than that of the launcher tube, some portion of the rocket will be inside the launcher tube and the remaining portion will be outside the launcher. So as per convenience Face split is done on the rocket at a length of 0.60515mm from the base of the motor tube another face splits done at $1/4^{th}$ length of the motor tube on its inner surface to apply temperature during analysis as shown in Fig 3.19.

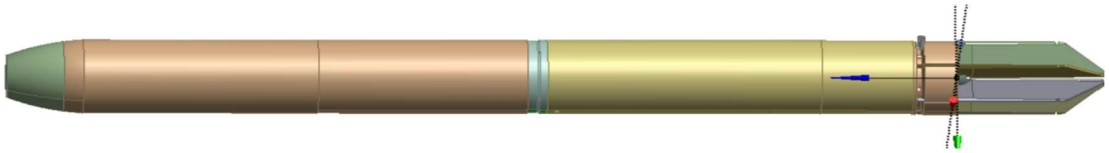


Fig 3.19 Geometry of Face splits

To create the frictionless contact between the Rocket and Launcher tube a face split with 10^0 Section is made on the outer surface of the Rocket as shown in Fig 3.20

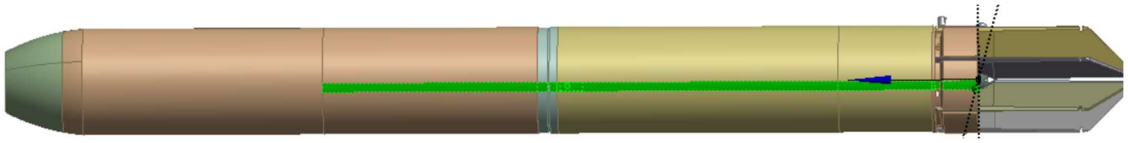


Fig 3.20 Geometry of Face split on the rocket surface

e. Contact generation

Contacts and contact behaviors generated by an interface generator are automatically associated with interface generator that created them. Every contact generated by an interface generator is assigned the same contact behavior if the "one shared by all contact" method is used to generate the contacts. Thus, if there is a need to modify the properties for this contact behavior, properties are modified to be attached to all contact pairs generated with that contact pair.

Since all the parts are connected with threaded joints, bonded contacts are used with Augmented Lagrange as Formulation Method.

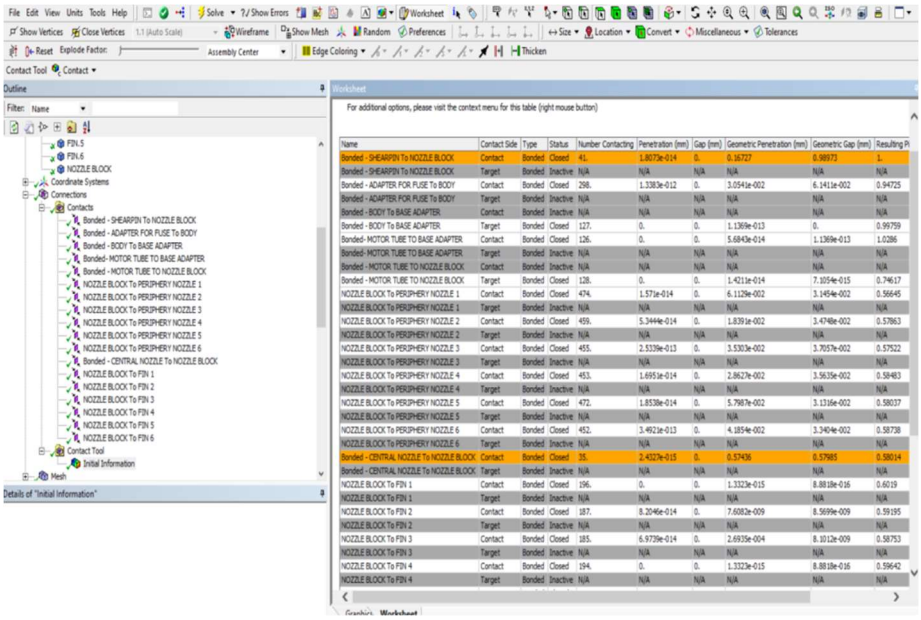


Fig 3.21 Contact Information

f. Mesh generation

Meshing is an integral part of the engineering simulation process where complex geometries are divided into simple elements that can be used as discrete local approximations of the larger domain. The mesh influences the accuracy, convergence and speed of the simulation. Furthermore, since meshing typically consumes a significant portion of the time it takes to get simulation results, the better and more automated the meshing tools, the faster and more accurate the solution.

Ansys provides general purpose, high-performance, automated, intelligent meshing software which produces the most appropriate mesh for accurate, efficient Multiphysics solutions from easy, automatic meshing to highly crafted mesh. Methods available cover the meshing spectrum of high-order to linear elements and fast tetrahedral and polyhedral to high-quality hexahedral [8]

In this project Hex dominant and tetrahedral mesh is used and refined by face sizing and body sizing. Meshed models of the rocket and the launcher are shown in Fig 3.22 and Fig 3.23 respectively.



Fig 3.22 Meshed Model of Rocket Assembly



Fig 3.23 Meshed Model of Launcher

3.3.2 Finite Element Model Weights of MRCR Rocket

Weight details of the rocket obtained from the finite Element Model are as shown in Table 3.2

Table 3.2 Finite Element Model Weights of MRCR Rocket

Component Details	**Weight in Kg**
Motor Tube	2.6278
Nozzle Block Assembly	2.034
Base Adapter	0.7206
Body	4.9426
Fuse Adapter	0.48578
6 Fins	0.4908
Empty Rocket Weight	8.1
Chaff	3.2
Total Rocket Weight	11.30

Weight Details given by CNAI, Indian Navy are shown in Table 3.3

Table 3.3 Weight Details given by CNAI

Weight of Rocket	8.1Kg
Weight of Pay load	3.2 Kg
Total Weight	11.30Kg

It is clear that the weights given by CNAI and weight of the Finite Element Model obtained are same, So Finite Element Model thus obtained is reliable

3.3.3 Thrust Details

Table 3.4 Thrust details given by CNAI

S. No	Type	Thrust in Kgf				Mass of propellant in Kg
		Average conditioning temperature			Maximum at all temperatures	
		$+27^0$C (12 hrs.)	-10^0C (12 hrs.)	$+50^0$C (12 hrs.)		
1	MRCR	560±50	500±50	580±50	≤ 1340	Min.2.65

To analyze issue-1 minimum thrust of 450Kgf and mass of propellant 2.65 kg is considered

$$450 \text{ kgf} = 450 \times 9.81$$

$$= 4414 \text{ N}$$

Therefore, thrust of 4414N is considered in the analysis.

CHAPTER 4 STEADY STATE THERMAL ANALYSIS

A steady state thermal analysis calculates a thermal response to specified heat loads subject to specified prescribed temperatures and/or convection conditions. It is used to determine temperatures, thermal gradients, heat flow rates and heat fluxes in an object that are caused by thermal loads that don't vary over time. Such loads include the following

- Convections
- Radiations
- Heat flow rates
- Heat flux (heat flow per unit area)
- Heat generation rates (heat flow per unit volume)
- Constant temperature boundaries

4.1 Problem Description

As discussed in the previous chapter one of the possible cause for issue-1 that is Rocket not leaving the launcher tube may be due to of temperature mismatch. As Rocket consists of fuel that burns at a very high temperature in order to provide sufficient thrust. Fuel that burns at higher temperature may also lead to thermal expansion of the rocket body as well as launcher tube. Initially the rocket rests in the launcher tube as shown in the figure. There is initial gap of 0.5mm between rocket and launcher tube. If the radial expansion of the rocket is more than that of launcher tube then there will be chance of rocket getting stuck in the Launcher tube So steady state thermal analysis is conducted to investigate the results.

4.2 Steady State Thermal Analysis of Rocket

4.2.1 CAD model

All parts are designed and assembled in CATIA V5 Software and imported to ANSYS Workbench. CAD Model of the rocket is shown below.

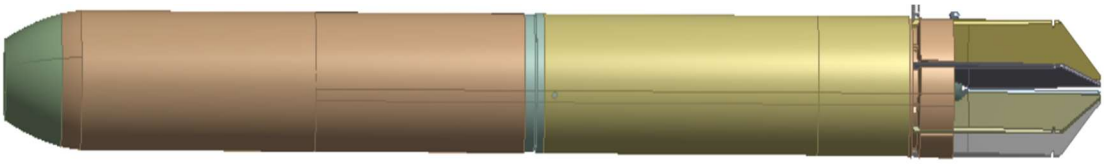


Fig 4.1 CAD Model of the Rocket Assembly

4.2.2 Mesh generation and Boundary Conditions

Hex dominant mesh is used for body and fuse adapter, tetrahedral mesh is used for the remaining parts of the assembly. Refinement of mesh is done by face sizing and body sizing of the parts. The meshed model is shown in the figure below.



Fig 4.2 Meshed Model of the rocket assembly

For steady state thermal analysis, the boundary conditions are taken as the temperature at Fuse adapter as 40^0C and the temperature at inner walls of motor tube up to $1/4^{th}$ of its length is assumed to be 1000^0C. (Since the motor tube contains the fuel that burns to provide thrust).

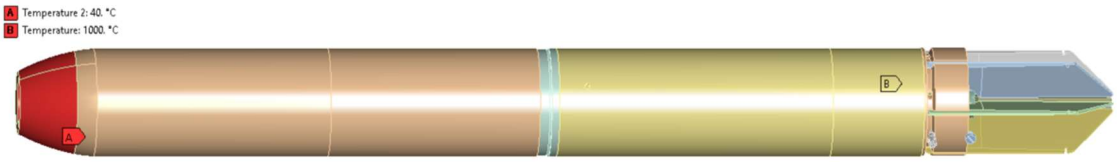


Fig 4.3 Assumed Thermal boundary conditions

In order to obtain the thermal stresses and thermal deformation, static structural analysis is done with the imported load as temperature distribution and the other boundary conditions as fixed supports at both the shear pin and stopper pin hubs since they are fitted in the launcher tube and frictionless support at the cross section where the rocket touches the launcher tube , the boundary conditions are shown in the Fig 4.4 **Error! Reference source not found.**,Fig 4.6.

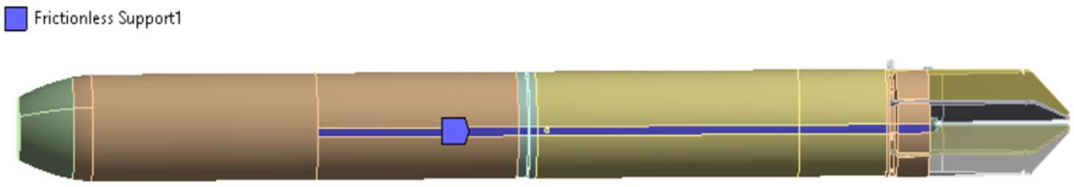


Fig 4.4 Frictionless support

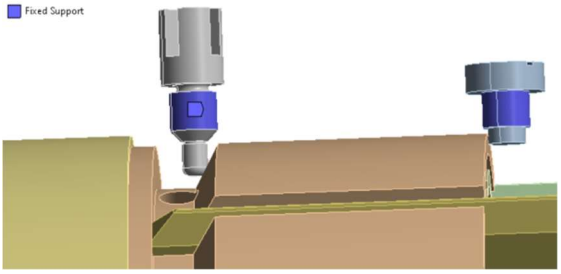


Fig 4.5 Fixed supports

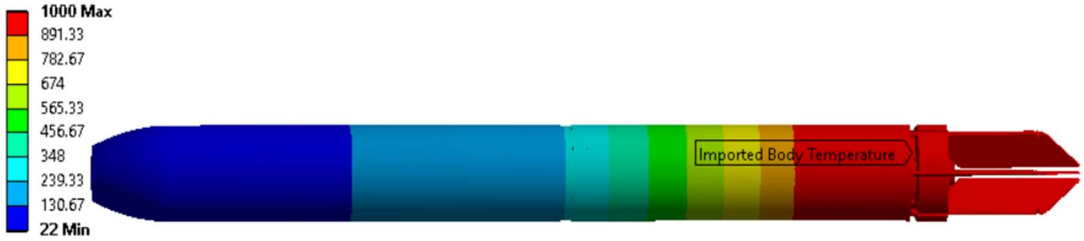


Fig 4.6 Imported body temperature of the rocket

4.2.3 Temperature distribution of rocket

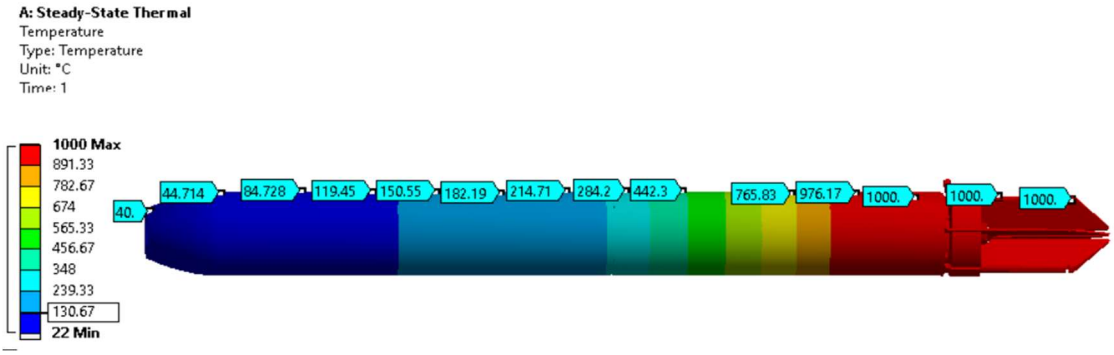


Fig 4.7 Temperature distribution of the rocket

4.2.4 Directional Deformation of the rocket

Since the rocket is placed in the launcher tube there will be no deformation on the bottom side where the rocket sits in the launcher tube. The directional deformation along the radial direction is shown in the Fig 4.8

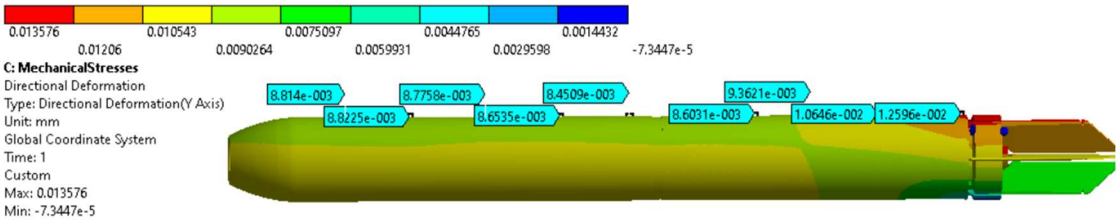


Fig 4.8 Directional deformation of rocket along the radial direction

4.3 Thermal Analysis of Launcher Tube

4.3.1 CAD model of Launcher tube



Fig 4.9 CAD Model of the Launcher Tube

4.3.2 Mesh generation and Boundary Conditions

Due to the irregularities in the geometry hex dominant mesh cannot be applied so tetrahedral mesh is applied.

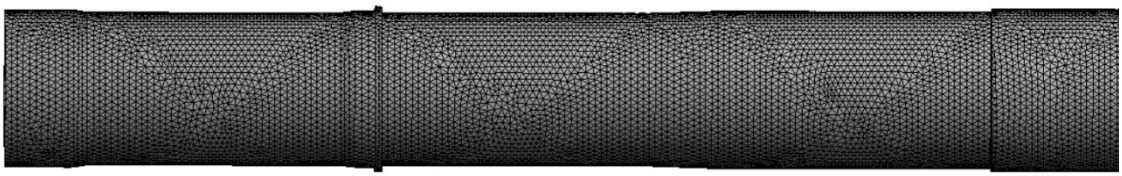


Fig 4.10 Meshed Model of the Launcher Tube

Temperatures that are obtained from the surface of the rocket are taken at four different locations by average and are applied at the corresponding lengths on the inner walls of the launcher tube. Boundary conditions applied are as shown in Fig 4.11

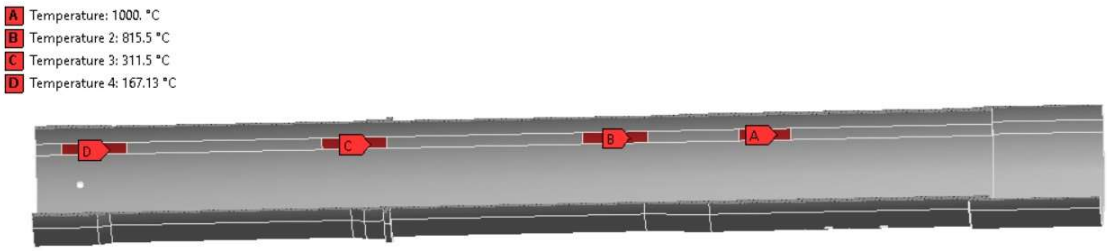


Fig 4.11 Sectional view of Launcher tube showing boundary conditions

Similarly, to obtain the thermal stresses and thermal deformation of the launcher tube static structural analysis is done with the imported load as temperature distribution and the other boundary conditions as fixed support at the shear pin hole and frictionless support at the bracket for front support as shown in Fig 4.12and Fig 4.13

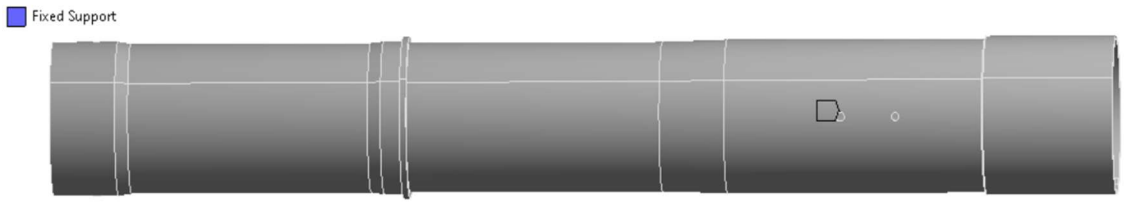


Fig 4.12 Fixed support at the shear pin hole

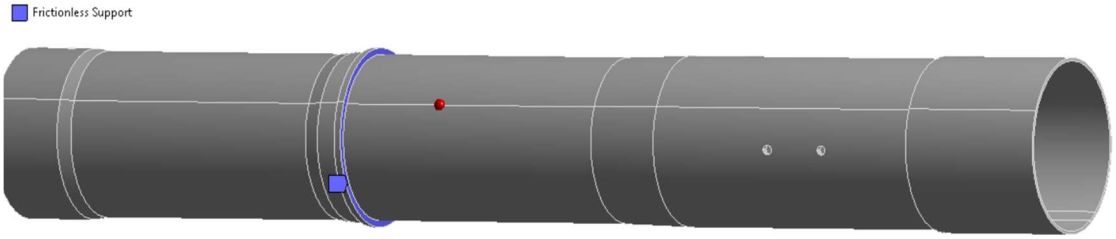


Fig 4.13 Frictionless support at bracket for support

4.3.3 Temperature distribution of Launcher tube

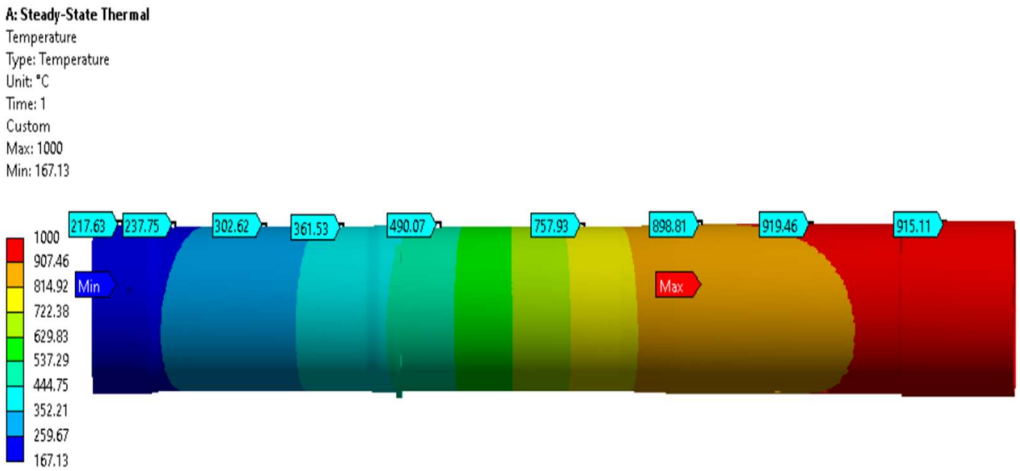


Fig 4.14 Temperature distribution of launcher tube

4.3.4 Directional Deformation of the rocket

As launcher tube is free to expand in the both the directions radially the sectional view of the radial deformation is shown in the Fig 4.15

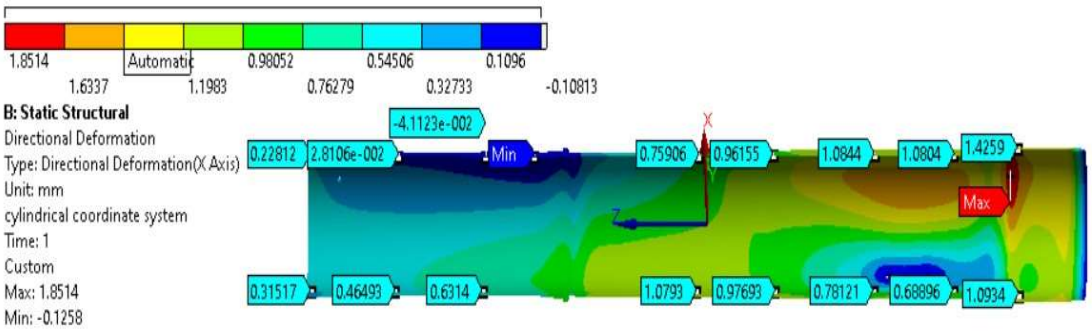


Fig 4.15 Directional deformation of Launcher tube along the radial direction

4.4 Comparison of Radial expansions of rocket and launcher tube

Initially, the gap between rocket and launcher tube is 0.5mm. By analysis, Radial expansion of Launcher Tube is more than that of the Rocket. The minimum gap between the rocket and the launcher tube is 0.98mm. Hence, Rocket doesn't get stuck in the Launcher Tube. Pictorial representation of comparison is shown in Fig 4.16

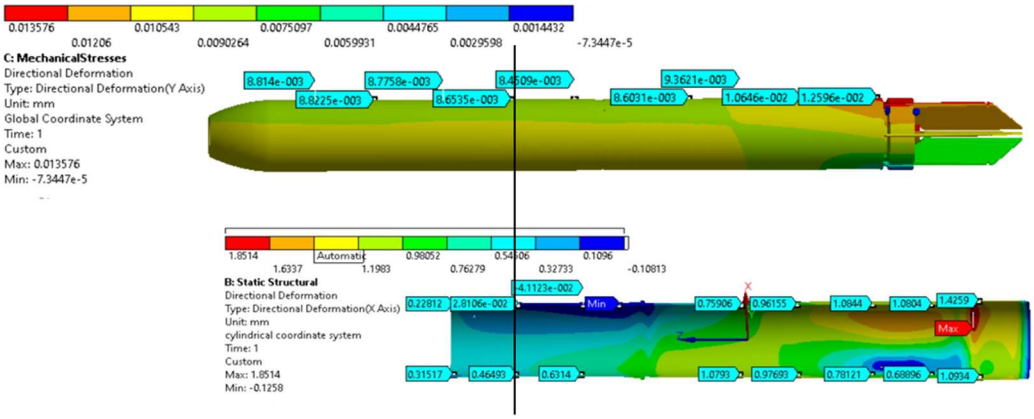


Fig 4.16 Comparison of Radial expansions of rocket and launcher tube

CHAPTER 5 STATIC STRUCTURAL ANALYSIS

The shear pin is inserted in the Nozzle block groove through its hole present on the launcher tube as shown in the Fig 5.1

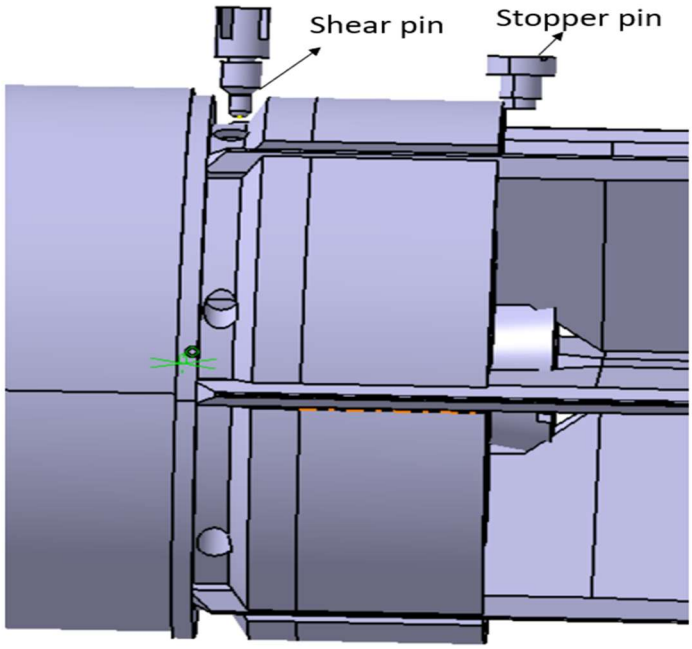


Fig 5.1 Position of shear pin in nozzle block groove

The main purpose of placing the shear pin in the Nozzle block groove is, it acts a fixed support until sufficient thrust is developed by the exhaust gases which are produced by combustion of propellant present in motor tube. When sufficient thrust is produced, then Shear pin gets sheared at its neck region, resulting in rocket leaving the Launcher tube.

5.1 Problem Description

As discussed in the 1st chapter one of the possible cause for issue-1 that is Rocket not leaving the launcher tube may be due to high strength of the shear pin. Though sufficient thrust is developed by proper selection of propellant, if the shear strength of shear pin is so high such that rocket is unable to shear the shear pin, then the rocket gets stuck in the launcher tube. Static structural analysis is conducted to investigate the results.

5.2 Modeling of Rocket-Launcher Assembly

The modeling of Rocket-Launcher Assembly is done by using CATIA V5 software. The 3D models each component is modeled and are assembled as shown in Fig 5.2

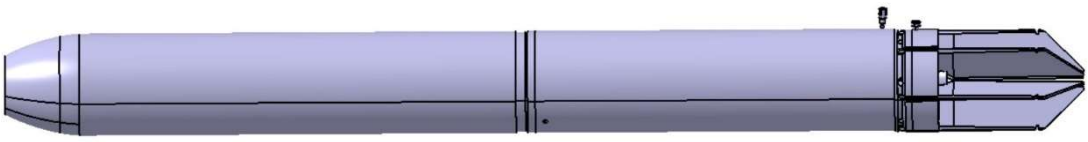


Fig 5.2 Rocket assembly with shear pin and stopper pin

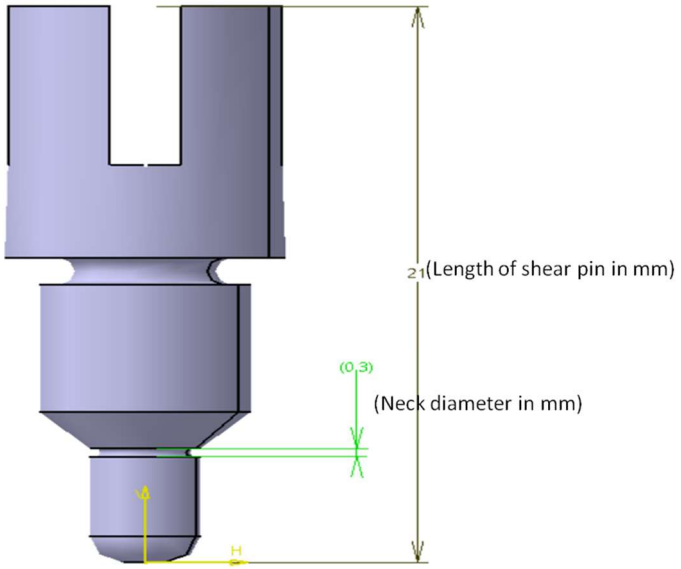


Fig 5.3 shear pin length and neck diameter

5.3 Static Structural Analysis of Rocket

The analysis is performed on rocket assembly with shear pin and stopper pin by using ANSYS software. A Static structural analysis determines the displacements, stresses, strains and forces in the structures or components caused by loads that do not induce significant inertia and damping effects. The 3D model of rocket launcher assembly in CATIA as shown in figure is converted into “step” file and then imported into geometry of static structural cell and then click on generate. This will generate the created CAD model into the design modeler window.

After importing the geometry and now define the materials properties of AA2014, AISI1040, AISI316, and AISI4340 in Engineering Data and assign the materials for respective rocket components. In the Model cell in analysis, contacts between rocket components are given. The generated model is now ready for meshing and static structural analysis.

5.3.1 Meshing and Boundary Condition

A hex dominant solid element is used in mesh generation of fuse adapter and body of rocket assembly and tetrahedron solid element is used in mesh generation of Base adapter, Motor tube, Nozzle block assembly, launcher tube, shear pin and stopper pin[10].



Fig 5.4 Meshed Model of the Rocket

In the Setup cell, boundary conditions such as loads and supports are defined Boundary conditions as fixed supports at both the shear pin and stopper pin hubs since they are fitted in the launcher tube and frictionless support at the cross section where the rocket touches the launcher tube, thrust application the boundary conditions are shown in Fig 5.5,Fig 5.6 and Fig 5.7

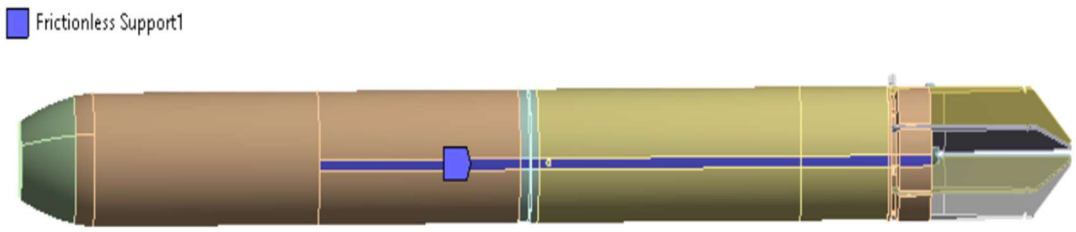


Fig 5.5 Frictionless support

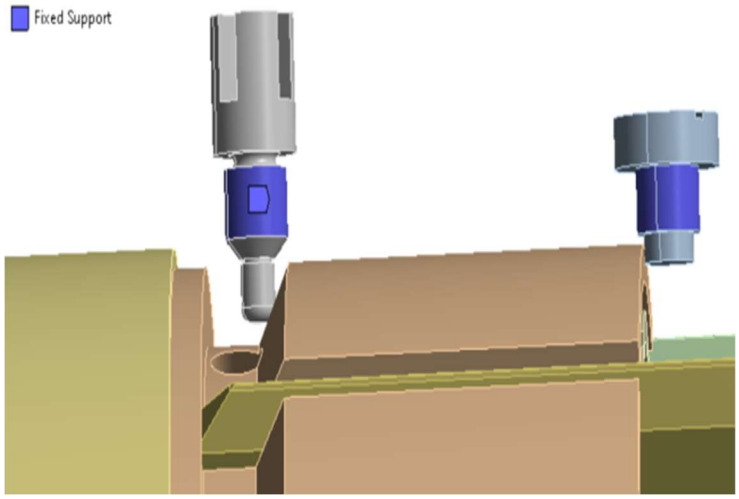


Fig 5.6 Fixed supports

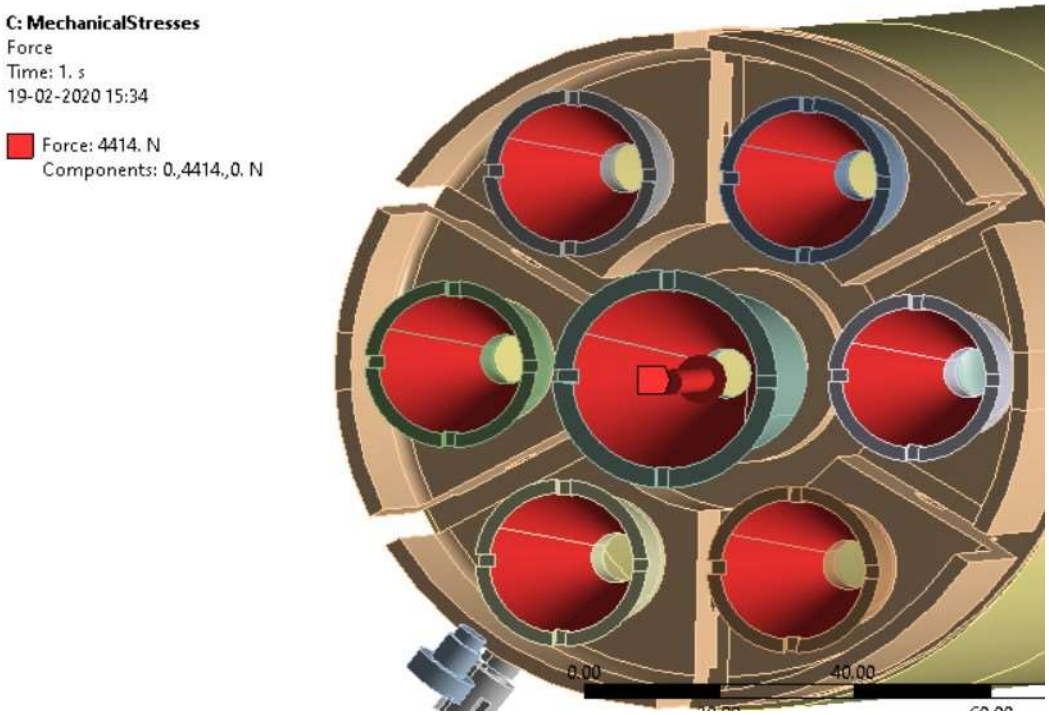


Fig 5.7 Thrust application in finite element model

5.4 Solution

After applying the boundary conditions on the model, by solving, the results of static structural analysis are obtained.

5.4.1 Mesh convergence study of shear pin

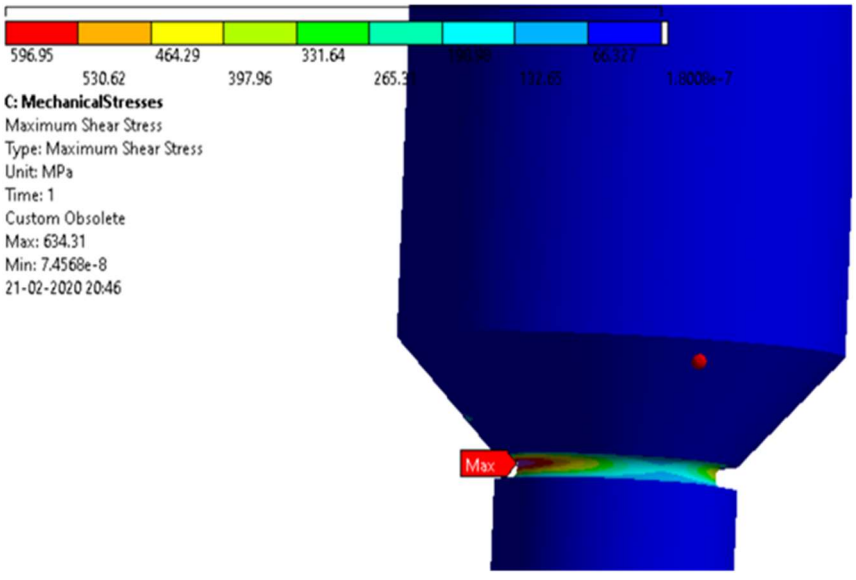


Fig 5.8 Maximum shear stress at neck region

Maximum shear stress mesh convergence(Shear Pin)	
Element size(mm)	**Maximum Shear stress(Mpa)**
0.08	596.95
0.06	614.64
0.05	626.34
0.03	631.53
0.02	634.35
0.01	634.52

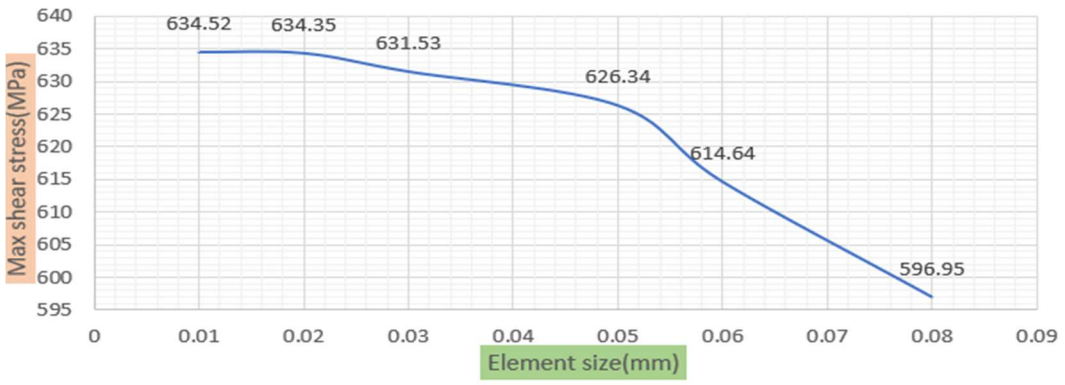


Fig 5.9 Maximum shear stress values at different sizes of mesh element

From the above analysis, shear stress values at neck region are observed. The maximum shear stress value is observed at neck region of the shear pin as shown in the Fig 5.8. The size of the mesh element at neck region is refined and a graph is plotted between maximum shear value and element size for different sizes of mesh element as shown in the Fig 5.9. It is observed that as element size decreases the maximum shear stress value increased and becomes stable as shown. Therefore, the size of the mesh element at neck region of shear pin is taken as 0.03mm.

5.4.2 Results

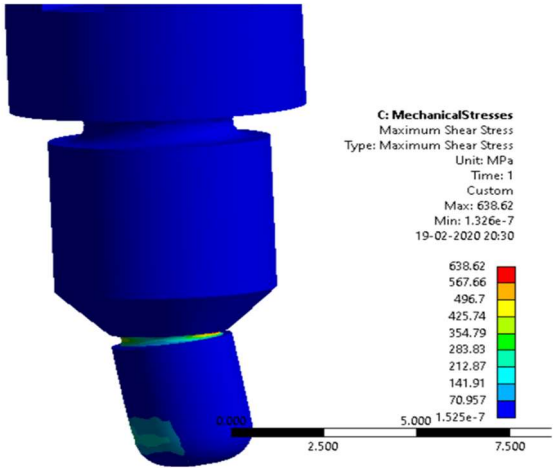


Fig 5.10 Shear stress contour in shear pin

Maximum shear stress observed in the neck region of the shear pin (638.62MPa) is higher than the shear strength of the EN24 material which is equivalent material for AISI 4340 (393MPa). From this it can concluded that, when sufficient thrust (4414N) is generated rocket assembly shears the shear pin and leaves the Launcher tube. Therefore, high strength of shear pin is not the possible cause for issue-1.

CHAPTER 6 TOLERANCE STACKUP ANALYSIS

Tolerance plays a vital role in any industrial product. Designer has to be very careful while dealing with tolerance. It is highly demanded to sell a product at low cost with higher accuracy. So it can perform efficiently. To fulfill such requirements tolerance stack up analysis is helpful.[12]

6.1 Tolerance stack-up condition

In a mechanical assembly, individual components are seldom produced in unique sizes. Their functional dimensions can always be produced within some tolerance due to manufacturing and other limitations. Thus for a given set of tolerances associated with individual dimension, the tolerance accumulated on the assembly dimension needs to be estimated. This is usually called tolerance analysis. The accumulated tolerance on the assembly dimension must be equal to or less than the corresponding assembly tolerance specified by the designer based on the functional and assembly requirements. In the tolerance techniques, the different criteria used for establishing relation between accumulated tolerance on the assembly dimensions and the assembly tolerance is popularly called as tolerance stack-up conditions. Over the years various researchers have proposed different models for determining stack-up conditions.

6.2 Problem Description

6.2.1 Assumption-Stopper pin locating in fin groove

The rocket assemblies are the heavily loaded into the launcher tube. There is a possibility of stopper pin tip locating itself into the nozzle block fin groove as, width of the nozzle block fin groove (W) is 5mm and diameter of the stopper tip (D) is 4.7 mm, if stopper pin tip can deform the triangular web of the fin.

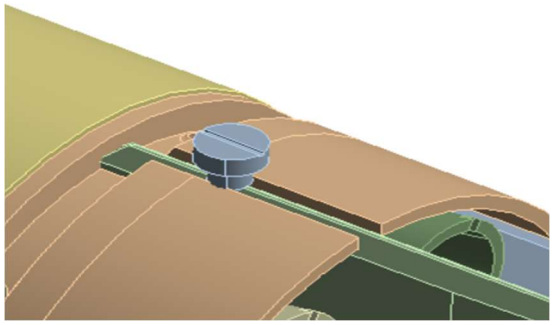


Fig 6.1 Stopper pin locating in fin groove

6.3 Tolerance stack-up calculations

To know whether the tip of the stopper pin can deform the triangular web of the fin, the thickness of the fin should be found at the depth where stopper pin tip penetrates into the fin.

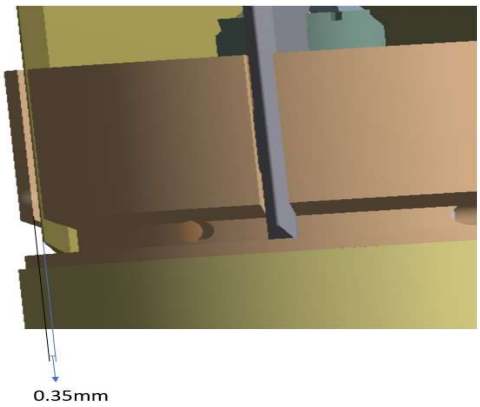


Fig 6.2 Distance between tip of the fin and nozzle block edge

When seen from the central axis of the rocket assembly, the tip of the fin is a lower than the nozzle block edge. Distance between tip of the fin and nozzle block edge is 0.35mm.

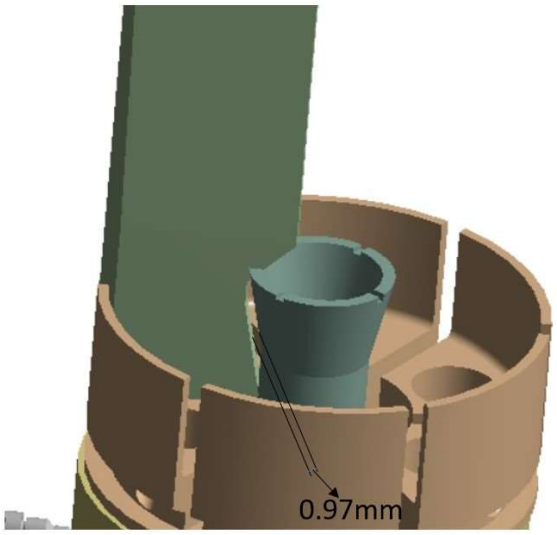


Fig 6.3 Maximum distance stopper pin can bend the fin

Distance between extreme end of central nozzle and extreme interior end of fin is 0.97mm. This distance can also be called as the maximum distance the stopper pin can bend the fin.

Length of the stopper pin tip on which rocket rests is 2mm.Considering the open tolerance according to the standards IS2012, Open tolerance for 2 mm is ±0.1mm.

New Length of the stopper pin tip (considering tolerance)

= 2+0.1

=2.1

Before the stopper pin tip to penetrate into the fin, it has to overcome the distance between tip of the fin and nozzle block edge as well as the distance up to which fin bends. Therefore, the length of the tip that penetrates into fin (L) is given by

L = (Original length of the tip) -(distance between tip of fin and nozzle block edge) -(distance up to which the fin bends)

= 2.0- (0.35)- (0.97)

= 0.77

The length of the stopper pin tip that penetrates into fin is 0.77mm. The thickness of the fin at depth of 0.77mm from its tip taking the taper angle of the fin @30°.

= tan (15) x0.77x2

= 0.41 mm.

Considering the open tolerance according to standards IS 2012, the tolerance for 30° is ±30'. The thickness of the fin at depth of 0.77 mm from the tip taking the angle of the fin @ 30° 30'.

= tan (15.25) x0.77x 2

=0.42 mm.

Table 6.1 The thickness of the fin at a depth 0f 0.77 mm from its tip.

Angle of the Fin	Thickness(mm)
30°	0.41
30°30'	0.42

As 0.42mm thickness is very small and material of the fin is aluminum, stopper pin tip can easily deform the triangular web of the fin and settle in gap between the fin and nozzle block fin groove. Then the shear pin gets located on the motor tube and there may be a possibility that shear pin may not get sheared, which in turn leads to rocket not leaving the launcher tube.

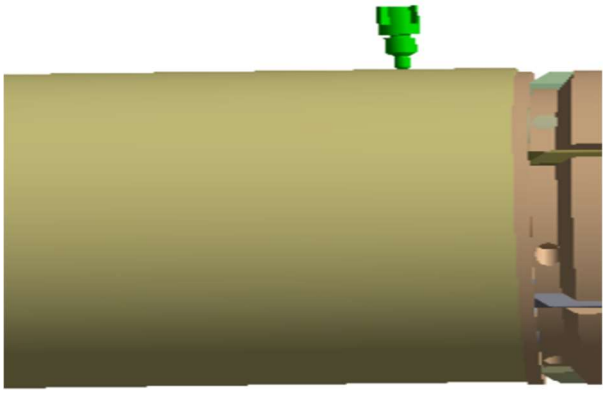


Fig 6.4 Shear pin located on motor tube

6.3.1 Position of stopper pin and shear pin in normal and tolerance stack up condition

In Kavach launching rocket system, the stopper pin is readily fitted in the launcher tube through its hole. The rocket assembly is loaded into the inclined launcher tube such that the nozzle block edge rests on the tip of the stopper pin and the shear pin is then inserted in the nozzle groove through its hole on launcher tube. In tolerance stack up condition the relative positions of the shear pin and stopper pin changes.

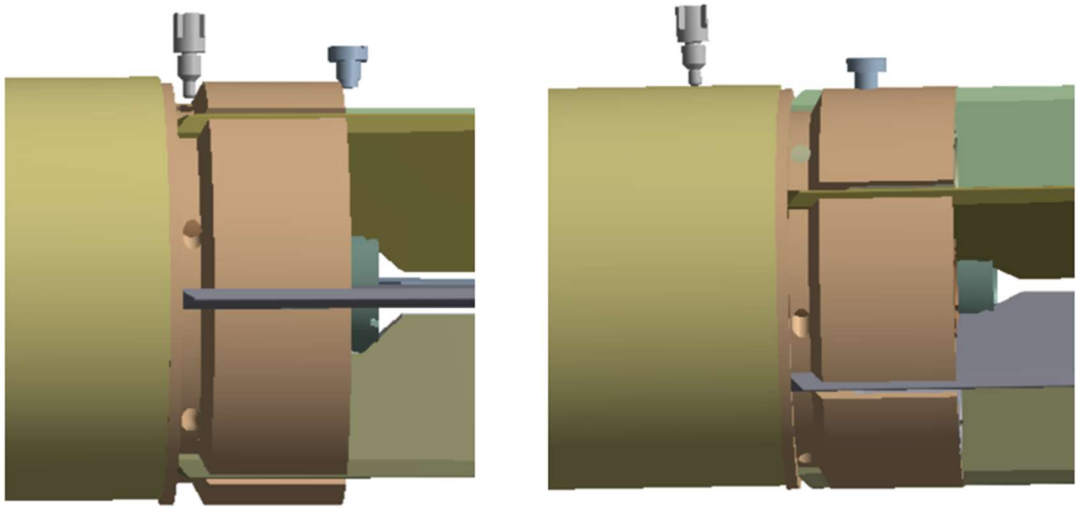


Fig 6.5 Position of stopper pin and shear pin in normal and tolerance stack up condition

6.4 Modeling of Rocket Assembly

The modeling of Rocket Assembly is done by using CATIA V5 software. The 3D models each component is modeled and are assembled as shown in the figure.

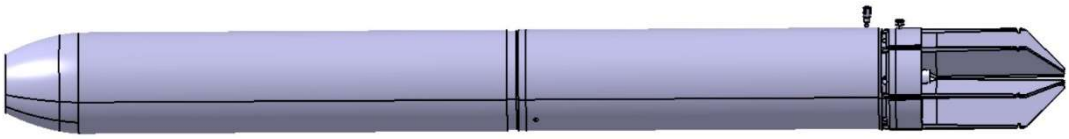


Fig 6.6 Assembly of Rocket in Tolerance stack up condition

6.5 Analysis to study stalling of rocket due to tightening shear pin on rocket body

The analysis is performed on rocket-launcher assembly by using ANSYS software. A Static structural analysis determines the displacements, stresses, strains and forces in the structures or components caused by loads that do not induce significant

inertia and damping effects. The 3D model of rocket launcher assembly in CATIA as shown in figure is converted into “step” file and then imported into geometry of static structural cell and then click on generate. This will generate the created CAD model into the design modeler window.[13]

After importing the geometry and now define the materials properties of AA2014, AISI1040, AISI316, and AISI4340 in Engineering Data and assign the materials for respective rocket components.

In the Model cell in analysis, contacts between rocket components are given. The generated model is now ready for meshing and static structural analysis.

6.5.1 Mesh generation and Boundary condition

A hexahedral solid element is used in mesh generation of fuse adapter and body of rocket assembly and tetrahedron solid element is used in mesh generation of Base adapter, Motor tube, Nozzle block assembly, launcher tube, shear pin and stopper pin.



Fig 6.7 The meshed model in ANSYS

In the Setup cell, boundary conditions such as loads and supports are defined.

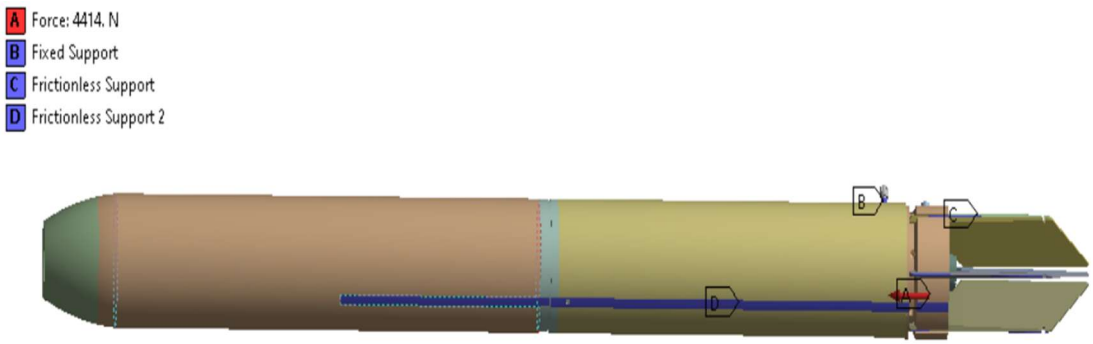


Fig 6.8 Application of boundary conditions

The initial boundary condition is force 4414N is given to all exhaust faces of the central and peripheral nozzles.

The second boundary condition is fixed support given to the faces of stopper pin and shear pin which come in contact with the launcher tube.

The third boundary condition is frictionless support which is given on all the tapered sides of the fins as they will be in contact with interior surface of launcher tube.

The fourth boundary condition is frictionless support which is given the exterior bottom slice of rocket assembly as it will rest on launcher tube.

6.6 Solution

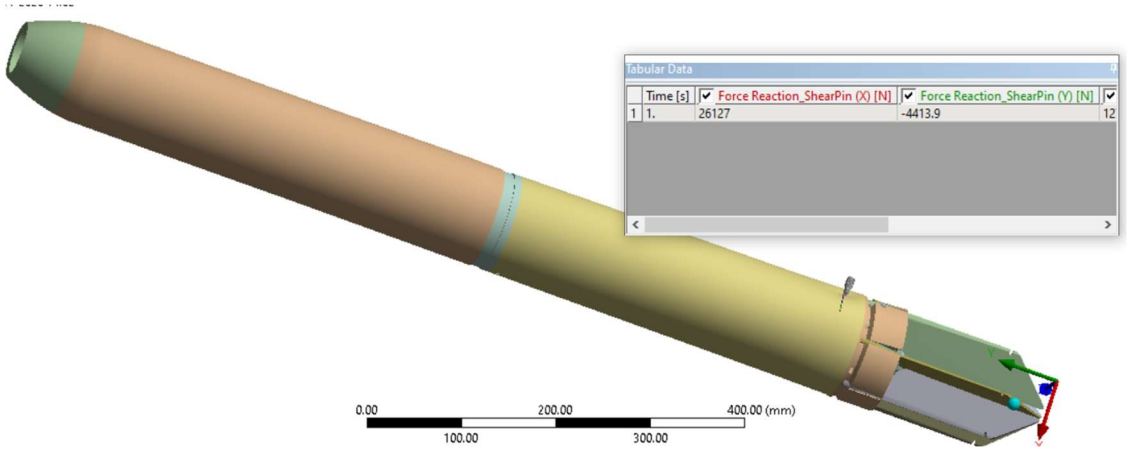


Fig 6.9 Reaction force on shear pin

After applying the boundary conditions on the model, by solving the results of static structural analysis are obtained. The reaction force at the shear pin is 4413.9 N.

So thrust force is capable to shear the shear pin. Therefore, the rocket leaves the launcher tube.

CHAPTER 7 THREAD JOINT ANALYSIS

Threaded parts are the most widely deployed mechanical joint in machinery design. The most common threads are symmetric ones, typically metric, Whitworth and Buttress. A key characteristic of a joint is the load distribution pattern through the threads along the depth path of the joint. This pattern helps when analyzing the joint's performance in some applications, especially if the material characteristics of the joint parts are rather different from those of the fastener part.

7.1 Problem Description

The fins are attached to the nozzle block, in the fin groove by means of spring support. The nozzle block with fins is fitted to the motor tube of rocket by a threaded joint.

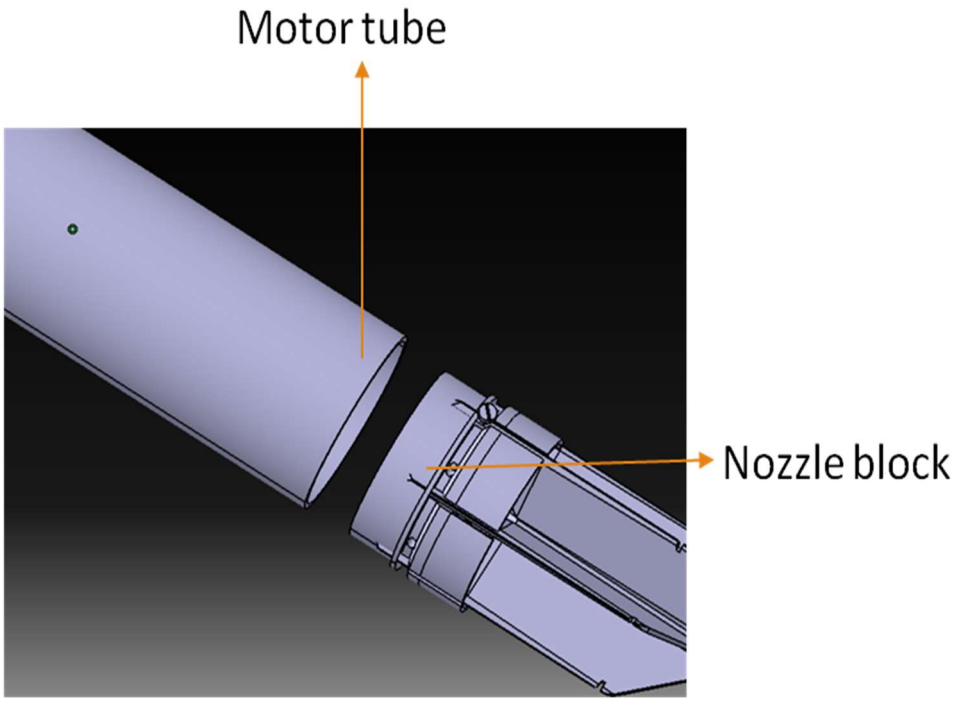


Fig 7.1 Motor tube and nozzle block

If the threaded joint between Motor tube and Nozzle block does not have sufficient thread contact length, there may be a chance of unthreading of the joint, thereby separation of Motor tube and Nozzle block. Therefore, there is a possibility of rocket leaves the launcher tube without fins.

7.2 Modeling Original Threaded joint between Motor tube and Nozzle block

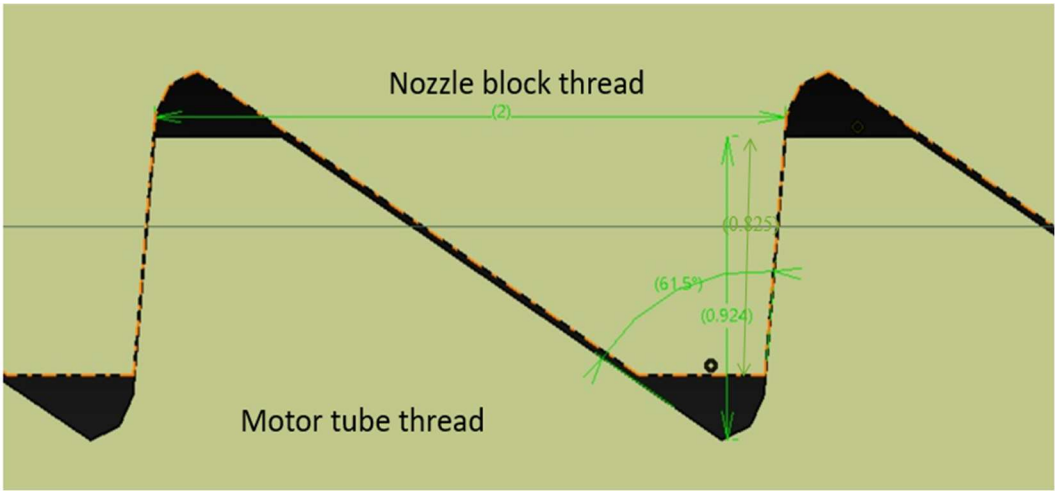


Fig 7.2 Original threaded joint between motor tube and nozzle block

The Original threaded joint between the Motor tube and Nozzle block is designed and assembled in CATIA V5 software as per dimensions given by CNAI. The above threaded joint shown in fig5.2 is Buttress thread joint of pitch 2mm and thread contact length (overlap) of 0.825mm.

7.3 Calculation of Overlap of Threaded Joint

Table 7.1 Calculations of Overlap of Threaded joint

OVERLAP AT NORMAL TEMPERATURE (Without Tolerance) (in mm)	
Internal Thread (Minor Diameter)	99.55
External Thread (Major Diameter)	101.2
Over lap	0.82
OVERLAP AT NORMAL TEMPERATURE (With Tolerance) (in mm)	
Internal Thread (Minor Diameter)	99.75
External Thread (Major Diameter)	101.05
Over lap	0.65
OVERLAP IN HOT CONDITION (Without Tolerance) (in mm)	
Internal Thread (Minor Diameter)	100.68
External Thread (Major Diameter)	102.40
Over lap	0.85
OVERLAP IN HOT CONDITION (Without Tolerance) (in mm)	
Internal Thread (Minor Diameter)	100.89
External Thread (Major Diameter)	102.25
Over lap	0.68

From the above calculations it is observed that, the threaded joint have the more root strength of the thread. The overlap of the threaded joint is increased in hot condition compared to the overlap in normal temperature for both the cases with tolerance and without tolerance. At the same time, the overlap of the threaded joint 0.825mm is much less than that of the overlap of standard buttress threads recommendation of **0.75P** i.e. 0.75 times 2=1.5 mm according to standards IS.4696.

Therefore, insufficient thread contact length may be the reason for rocket leaving the launcher tube without fins.

7.4 Designing and Modeling of New Modified Threaded joint between Motor tube and Nozzle block

The thread contact length of original threaded joint 0.825mm is less than suggested value of the standards IS 4696 i.e. 1.5mm. In order to match the thread contact length to suggested standard value, without compromising on the root strength value of the thread, the thread contact length is increased by 0.4mm.

New thread contact length = original thread contact length + (0.4)

= 0.825 + 0.4

= 1.225mm.

As the overlap of the threaded joint changes, the major and minor diameters of nozzle block and motor tube are also changed as shown in Table 7.2

Table 7.2 Major and minor diameters of nozzle block and motor tube for modified threaded joint

Contents	Major diameter(mm)	Minor diameter(mm)
Nozzle block	101.6	98.9
Motor tube	101.8	99.15

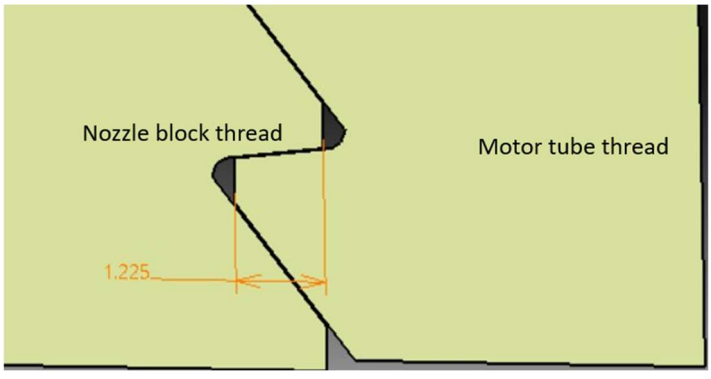


Fig 7.3 New modified threaded joint modeled in CATIA

The new modified threaded joint is modeled and assembled using CATIA V5 software with new overlap of 1.225mm.

7.5 Two dimensional axisymmetric analysis of threaded joints

To build reference model for our studies, axisymmetric geometry of the threaded joints is used. The main objective of this analysis is to focus on the Equivalent stresses on increasing the thread contact length of the threaded joint. The analysis is performed on both threaded joints by using ANSYS software. The threaded joints modeled and assembled in CATIA as shown in Fig 7.2, Fig 7.3 is converted into "step" files and then imported into geometry of static structural cell separately and then click on generate. This will generate axisymmetric model into the design modeler window.

After importing the geometry and now define the materials properties of AISI1040 and AISI 4340 in Engineering Data and assign the materials respectively to Nozzle block thread and Motor tube thread. The generated model is now ready for meshing and static structural analysis.

7.5.1 Mesh generation and Boundary condition

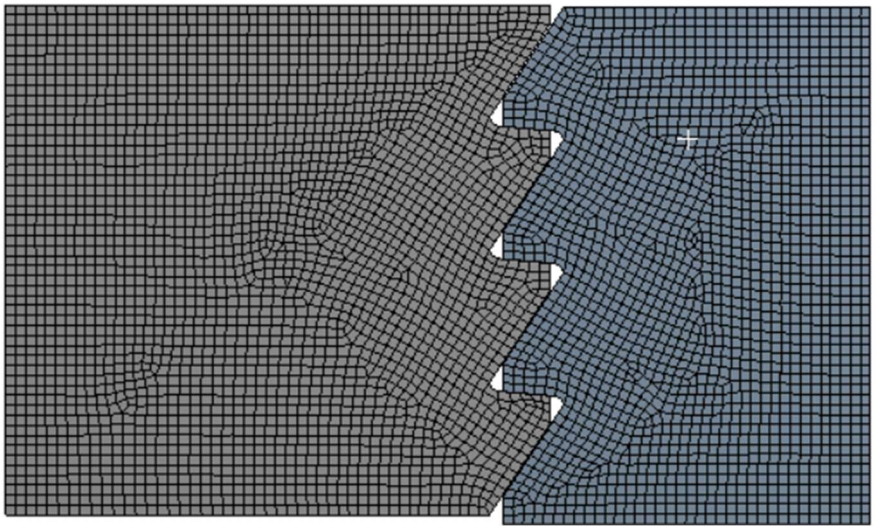


Fig 7.4 Meshed model of original threaded joint

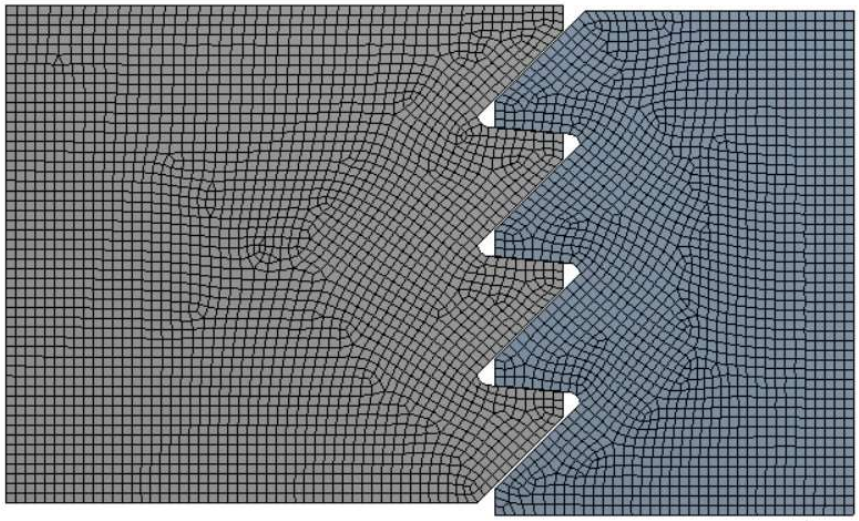


Fig 7.5 Meshed model of new modified threaded joint

A quadrilateral element is used in mesh generation of the threaded joints as shown in the Fig 7.5 and Fig 7.6. After meshing is done, in the Setup cell, boundary conditions such as loads and supports are defined.

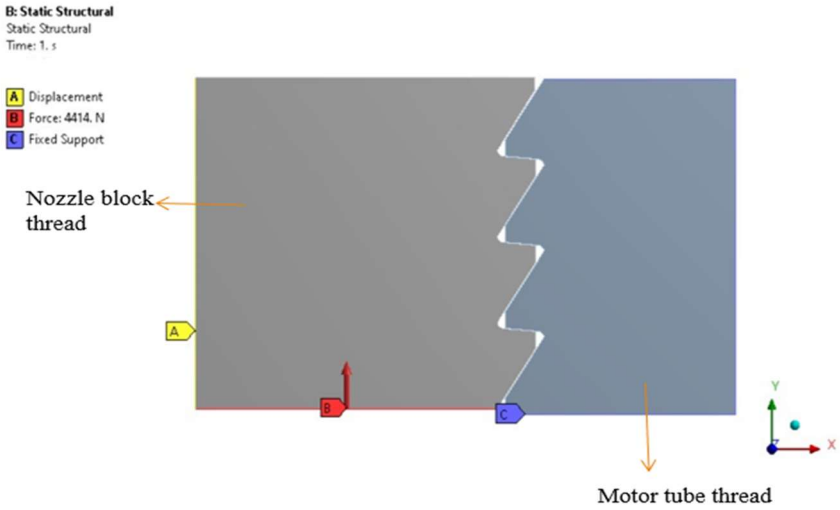


Fig 7.6 Application of boundary Conditions

The initial boundary condition, displacement is given alongside of nozzle block thread only in vertical direction as free (Y-=free) and horizontal direction (X=0).The second boundary condition, force is given to one edge of the nozzle block thread as thrust produced by nozzles is transferred to nozzle block. The direction of force is given such that the force is not transmitted along inclined sides. The third boundary condition, fixed support is given on all the three sides of the motor tube thread as it is fixed.

7.6 Solution

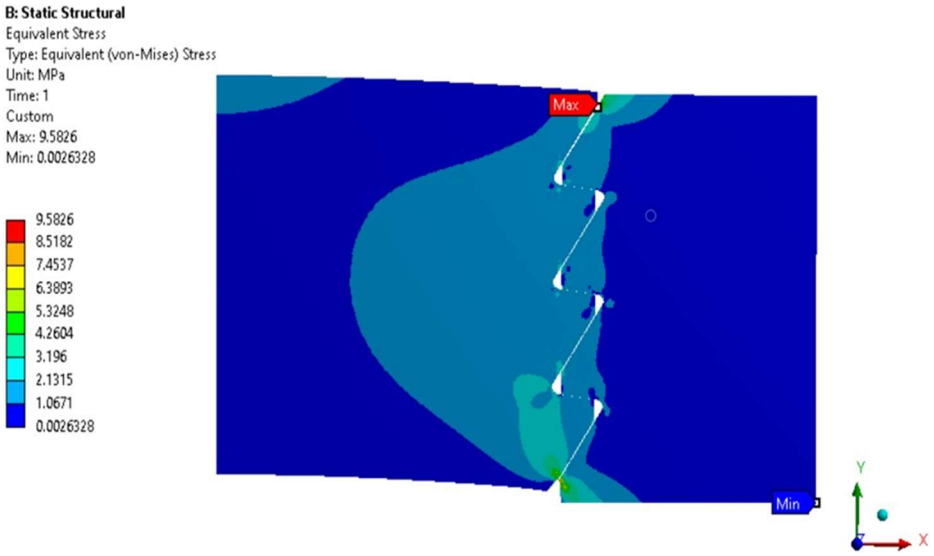


Fig 7.7 Equivalent stress for original threaded joint

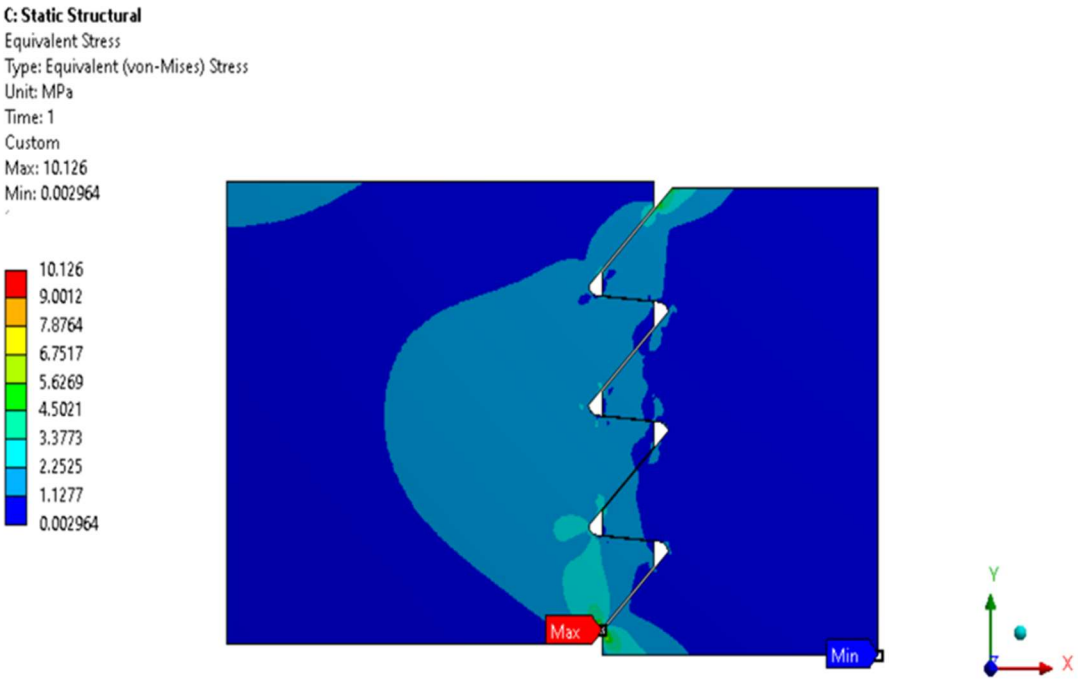


Fig 7.8 Equivalent stress for new modified threaded joint

Maximum equivalent stress of new threaded joint is more than that of the original threaded joint**.** Though the stress is increased, it is only very minor=10.12-9.58=0.54MPa. This slight increase has no major impact as the yield strength of the material is very high.

Yield strength of Nozzle block (AISI 1040) = 415MPa.

Yield strength of Motor tube (AISI 316) = 205MPa.

Therefore, by this analysis it can be concluded that thread contact length is increased without compromising in root strength of the thread. As the thread contact length is more, there will be no unthreading of threaded joint between nozzle block and motor tube. Now, the rocket leaves the launcher with fins.

CHAPTER 8 CONCLUSIONS

8.1 Conclusions

- The diameter of the stopper pin should be more than 5.3mm i.e., more than the width of the nozzle groove and considering the worst-case tolerance on nozzle block gap and stopper pin, this would still prevent the stopper pin from getting into the groove.
- The length of the stopper pin tip, where the nozzle block rests should be increased by 1.5mm so that it does not collide with the peripheral nozzles
- The overlap of the threaded joint should be increased in order to increase the strength of the threaded joint. Overlap of old and new threaded joints are shown in the **Error! Reference source not found.**

Table 8.1 Overlap for original and new modified threaded joints

Contents	Overlap(mm)
Original Threaded joint	0.825
New Modified Threaded joint	1.225

- Similarly, the major and minor diameters of nozzle block and motor tube are also to be changed as shown in Table 8.2

Table 8.2 Original and New dimensions

Contents	Original Dimensions (in mm)		New Dimensions (in mm)	
	Major diameter	Minor diameter	Major diameter	Minor diameter
Nozzle Block	101.2	99.3	101.6	98.9
Motor Tube	101.4	99.55	101.8	99.15

8.2 Future Scope of Study

- Full transient thermal analysis can be carried out to find actual temperature the motor tube sees instead of 1000^0C which is used temperature.
- Simulation of thread joints with manufacturing tolerance can be carried out

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